

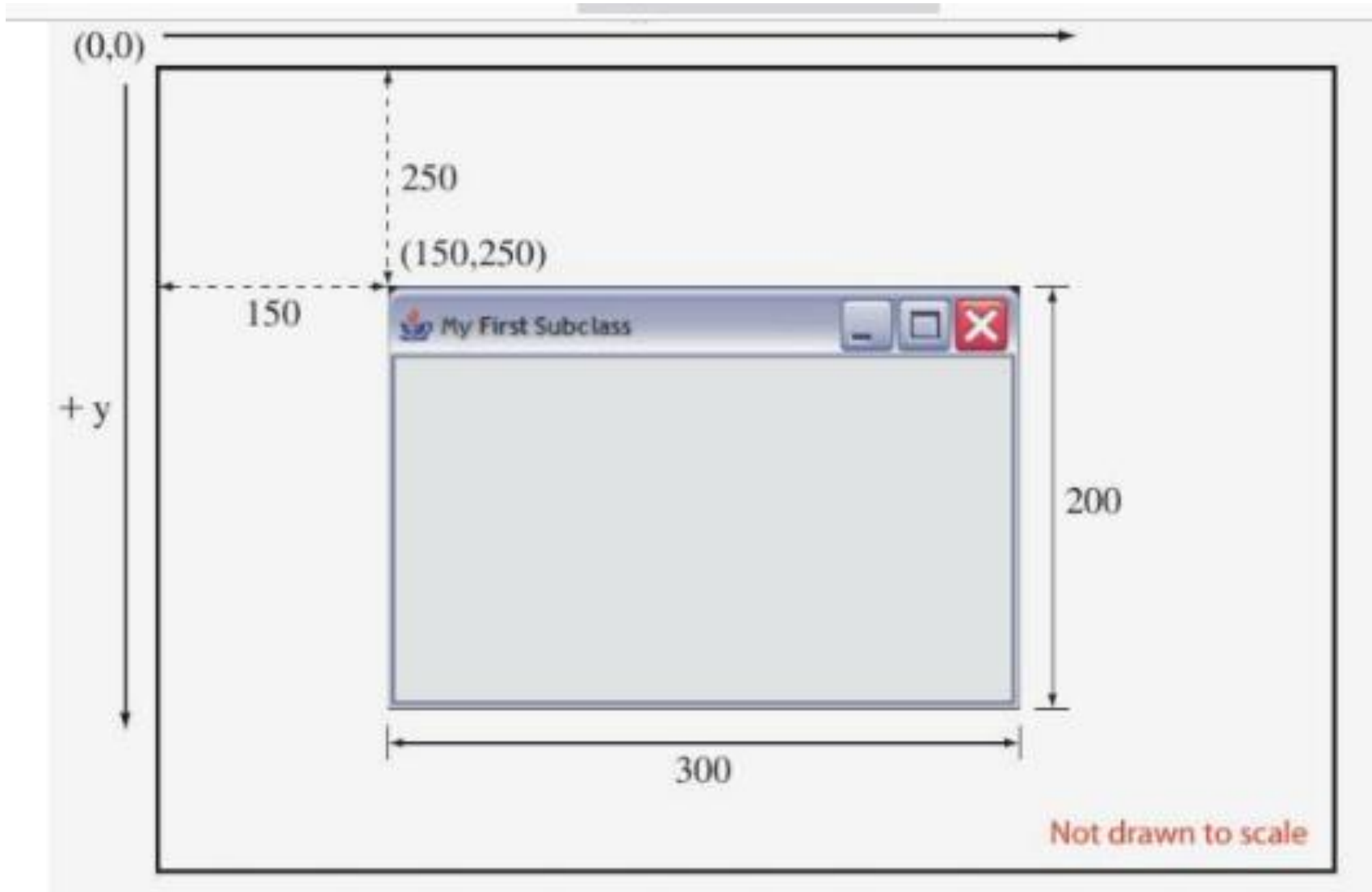


Programação Orientada a Objetos II

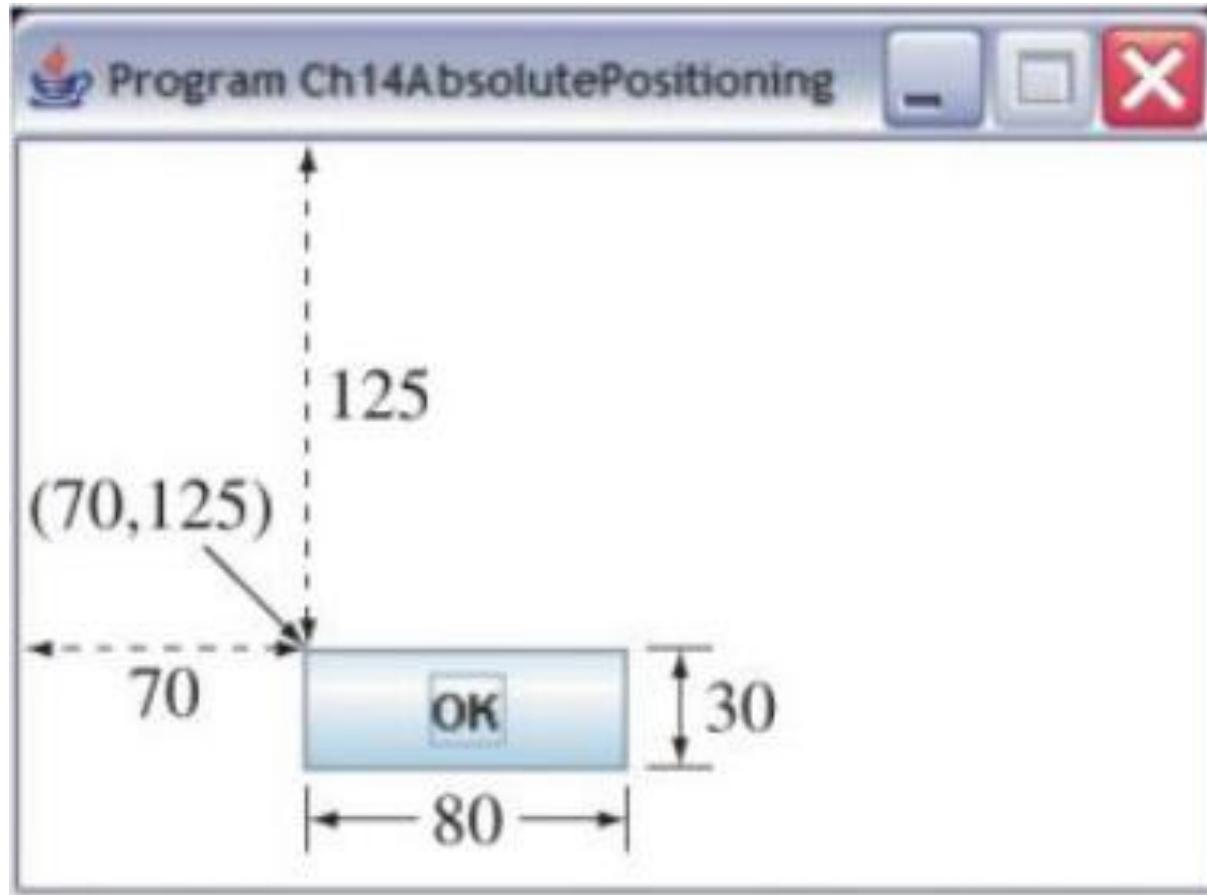
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Contato: angelaabreu@gmail.com

Interface Gráfica - SWING



Interface Gráfica - SWING



Tratamento de Eventos

Quadro 1.7 | Eventos na linguagem Java

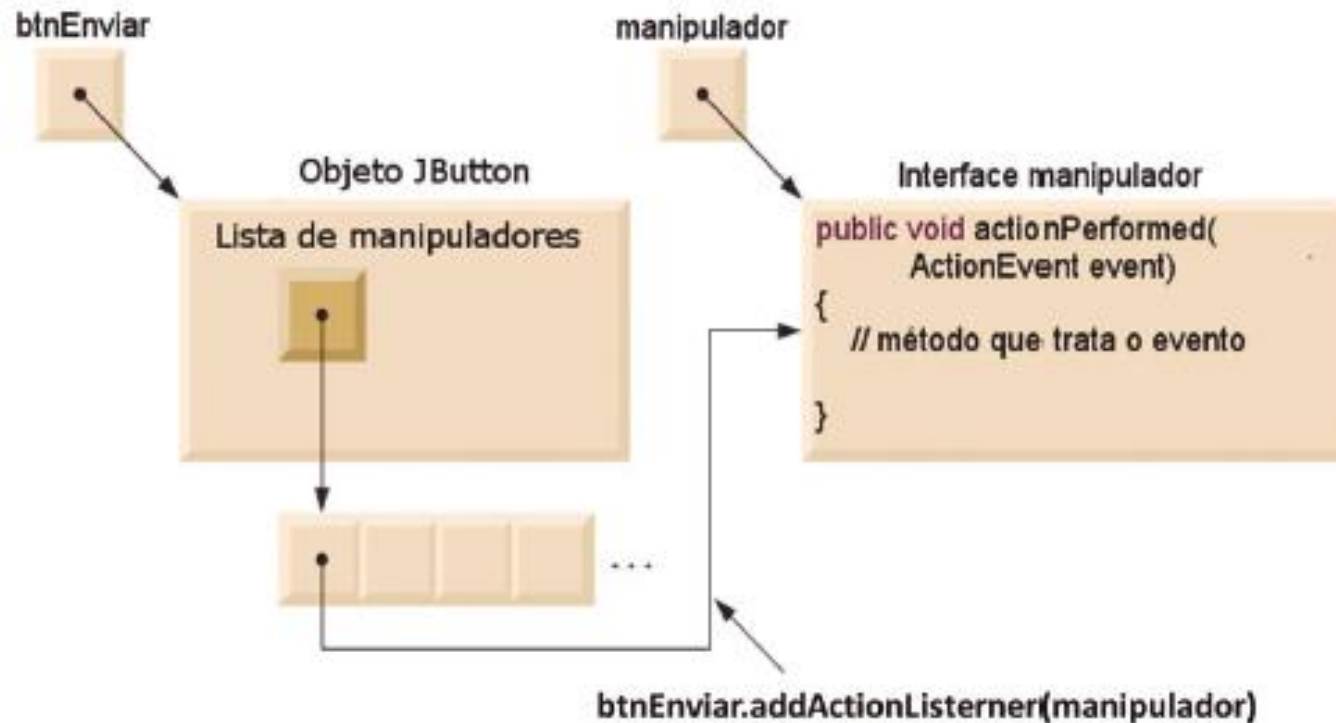


Evento	Descrição
ActionListener	Evento gerado pelo clique com o mouse em um botão.
ItemListener	Evento gerado quando um item em uma lista é selecionado.
FocusListener	Evento gerado quando um elemento ganha ou perde foco.
WindowListener	Evento gerado quando ocorre uma mudança na janela.

Fonte: elaborado pelo autor.

Interfaces

Figura 1.5 | Exemplo de como tratar o evento de clique em JButton do Java Swing

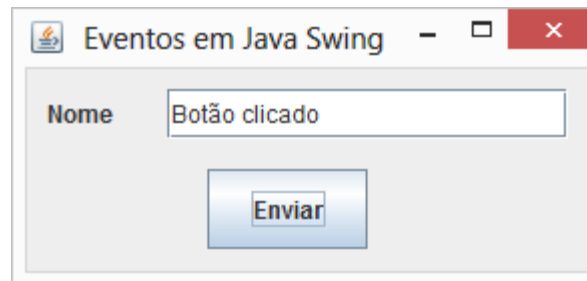
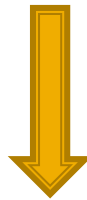
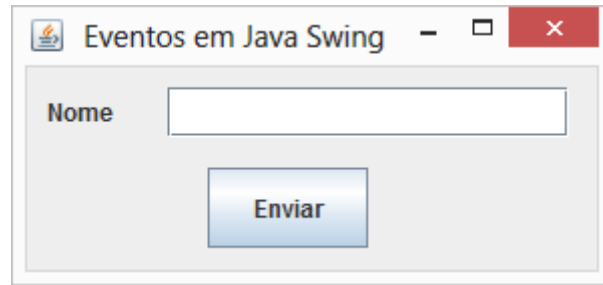


Fonte: adaptada de Deitel e Deitel (2016, p. 494).

Para implementar o tratamento de **eventos é necessário que a classe que está especializando o JFrame** tenha **alguma implementação dos métodos** da interface que trate os eventos e associe essa interface aos componentes que enviarão os eventos

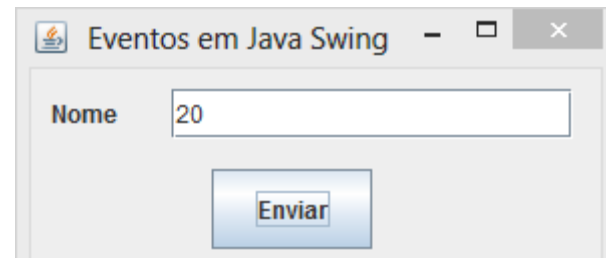
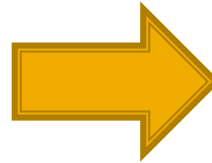
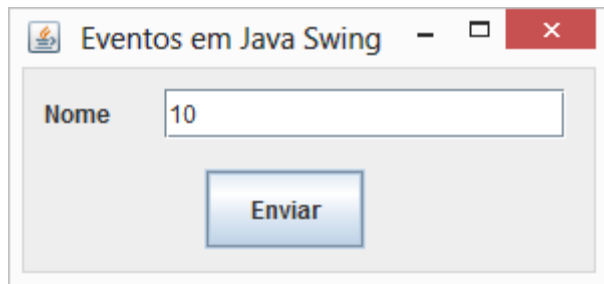
Exemplo

Implementar o evento abaixo utilizando **o método 1**: implementa a interface `ActionListener` diretamente na classe que especializa o `JFrame`

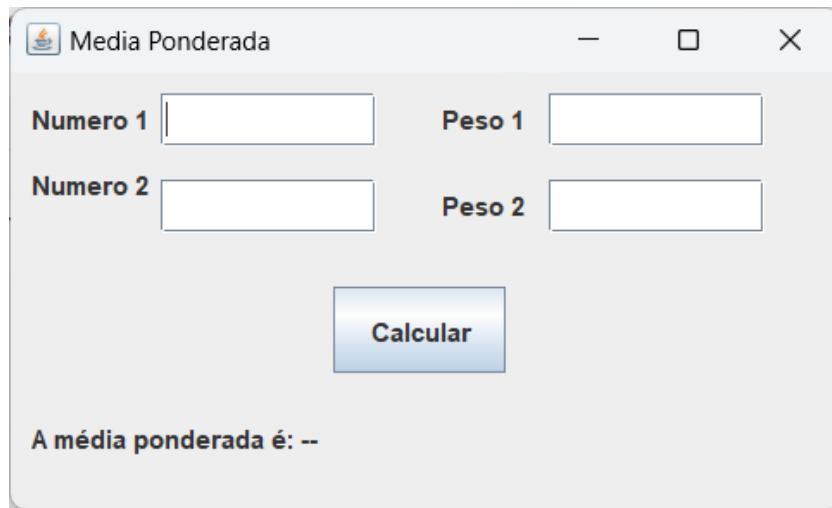


Exemplo

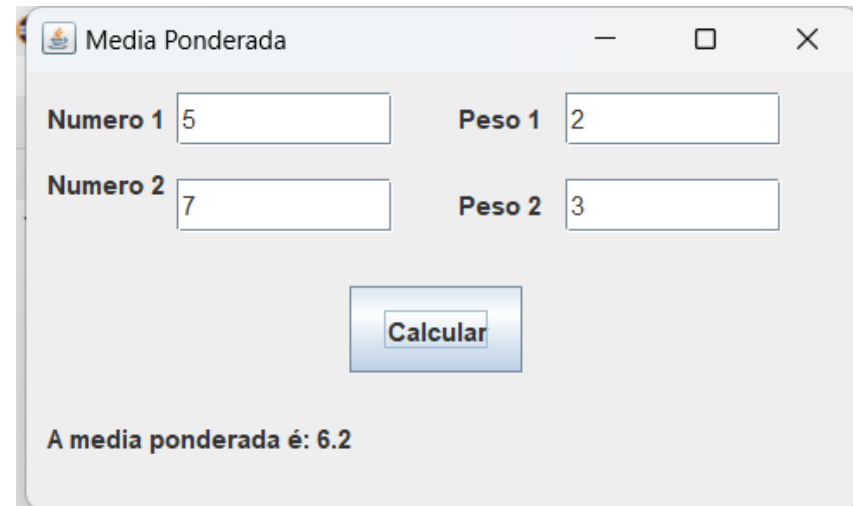
Implementar o evento abaixo utilizando **o método 2**: criar classe anônima que implementa a interface ActionListener



Exemplo



The screenshot shows a window titled "Media Ponderada" with standard Windows window controls (minimize, maximize, close). It contains four input fields: "Numero 1", "Peso 1", "Numero 2", and "Peso 2". All fields are empty. Below the input fields is a blue button labeled "Calcular". At the bottom of the window, the text "A média ponderada é: --" is displayed.



The screenshot shows the same "Media Ponderada" window after calculation. The input fields now contain the values: "Numero 1" is 5, "Peso 1" is 2, "Numero 2" is 7, and "Peso 2" is 3. The "Calcular" button remains visible. At the bottom, the text has changed to "A media ponderada é: 6.2".

$$\text{Média Ponderada} = \frac{(\text{Numero1} * \text{Peso1}) + (\text{Numero2} * \text{Peso2})}{(\text{Peso 1} + \text{Peso2})}$$

```
@Override
public void actionPerformed(ActionEvent e) {

    if(e.getActionCommand().equals("Calcular"))
    {

        float num1 = Float.parseFloat(txtNum1.getText());

        float num2 = Float.parseFloat(txtNum2.getText());

        float peso1 = Float.parseFloat(txtPeso1.getText());

        float peso2 = Float.parseFloat(txtPeso2.getText());

        float media = ((num1 * peso1) + (num2 * peso2)) / (peso1 + peso2);

        lblMediaPonderada.setText("A media ponderada é: " + Float.toString(media));

    }

}
```

Material Didático



Programação Orientada a Objetos II

Sumário

Unidade 1 Programação orientada a eventos com interfaces gráficas e banco de dados relacional	7
Seção 1.1 - Desenvolvimento de interfaces gráficas na linguagem Java	9
Seção 1.2 - Programação em Java usando orientação a eventos	27
Seção 1.3 - Programação em Java usando banco de dados relacional	47
Unidade 2 Programação concorrente orientada a objetos	67
Seção 2.1 - Programação em Java usando threads	69
Seção 2.2 - Definição e tratamento de exceções para sistemas com threads	89
Seção 2.3 - Programação em Java utilizando elementos para sincronização em Java	107
Unidade 3 Padrões de projeto, ferramentas e métodos ágeis	127
Seção 3.1 - Ferramentas para programação em linguagens orientadas a objetos	129
Seção 3.2 - Padrões de projetos em orientação a objetos	149
Seção 3.3 - Métodos ágeis em orientação a objetos	167
Unidade 4 Novas tecnologias para programação em banco de dados	183
Seção 4.1 - Banco de dados NoSQL	185
Seção 4.2 - Introdução ao desenvolvimento em Java usando MongoDB	200
Seção 4.3 - Desenvolvimento em Java usando MongoDB	216

PROVA URNA ELETR... — □ ×

Candidatos

- 1 - Joaquim Santos
- 2 - Karina Consta
- 3 - Voto em Branco

Voto:

Votar

Apuracao dos votos

Total de votos --

Total em branco --

Votos em candidato 1 --

Votos em candidato 2 --



URNA ELETRONICA — □ ×

Candidatos

- 1 - Joaquim Santos
- 2 - Karina Consta
- 3 - Voto em Branco

Voto:

Votar

Apuracao dos votos

Total de votos 3

Total em branco 0

Votos em candidato 1 1

Votos em candidato 2 2

Modelagem

PROVA URNA ELETR...

Candidatos

- 1 - Joaquim Santos
- 2 - Karina Consta
- 3 - Voto em Branco

Voto:

Votar

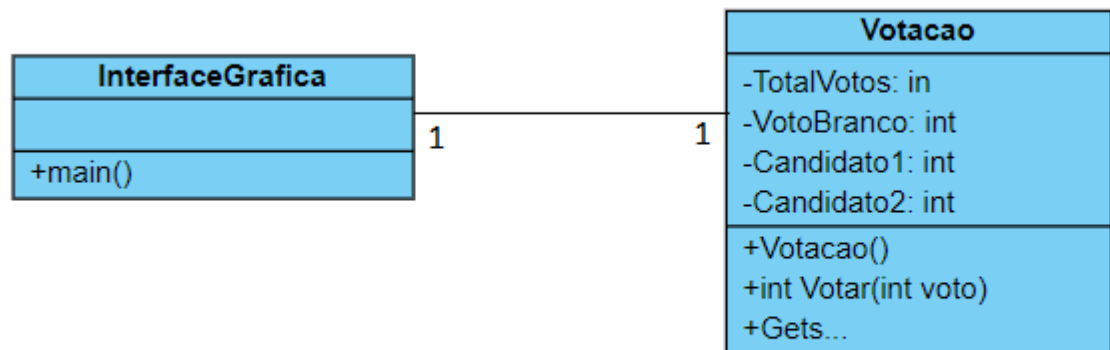
Apuracao dos votos

Total de votos --

Total em branco --

Votos em candidato 1 --

Votos em candidato 2 --



New Java Class

Java Class



 The use of the default package is discouraged.

Source folder: Exercicio01/src

Browse...

Package: (default)

Browse...

☐ Enclosing type:

Browse...

Name: Votacao

Modifiers:

☒ public ☐ package ☐ private ☐ protected

☐ abstract ☐ final ☐ static

☒ none ☐ sealed ☐ non-sealed ☐ final

Superclass: java.lang.Object

Browse...

Interfaces:

Add...

Remove

Which method stubs would you like to create?

☐ public static void main(String[] args)

☐ Constructors from superclass

☒ Inherited abstract methods

Do you want to add comments? (Configure templates and default value [here](#))

☐ Generate comments



Finish

Cancel

```
public class Votacao {
```

```
    private int Candidato1;  
    private int Candidato2;  
    private int VotoBranco;  
    private int TotalVotos;
```

```
    public Votacao()
```

```
    {  
        Candidato1 = 0;  
        Candidato2 = 0;  
        VotoBranco = 0;  
        TotalVotos = 0;  
    }
```

```
    public int Votar(int Voto)
```

```
    {  
        switch(Voto)  
        {  
            case 1: Candidato1++;  
                    break;  
            case 2: Candidato2++;  
                    break;  
            case 3: VotoBranco++;  
                    break;  
  
            default: return -1;  
        }  
  
        TotalVotos++;  
  
        return 1;  
    }
```

```
public int getCandidato1() {  
    return Candidato1;  
}
```

```
public int getCandidato2() {  
    return Candidato2;  
}
```

```
public int getVotoBranco() {  
    return VotoBranco;  
}
```

```
public int getTotalVotos() {  
    return TotalVotos;  
}
```

```
}
```

New Java Class

Java Class

Source folder: Exercicio01/src Browse...

Package: (default) Browse...

☐ Enclosing type: Votacao Browse...

Name: InterfacePrincipal

Modifiers: ☒ public ☐ package ☐ private ☐ protected
☐ abstract ☐ final ☐ static
☒ none ☐ sealed ☐ non-sealed ☐ final

Superclass: javax.swing.JFrame Browse...

Interfaces: java.awt.event.ActionListener Add... Remove

Which method stubs would you like to create?

☒ public static void main(String[] args)
☐ Constructors from superclass
☒ Inherited abstract methods

Do you want to add comments? (Configure templates and default value [here](#))
☐ Generate comments

Finish Cancel

```
1 import java.awt.Container;
2 import java.awt.event.ActionEvent;
3 import java.awt.event.ActionListener;
4
5 import javax.swing.JButton;
6 import javax.swing.JFrame;
7 import javax.swing.JLabel;
8 import javax.swing.JTextField;
9
10 public class InterfacePrincipal extends JFrame implements ActionListener {
11
12     private JLabel lblCandidato;
13     private JLabel lblTotal;
14     private JLabel lblTotalBranco;
15     private JLabel lblCandidato1;
16     private JLabel lblCandidato2;
17
18     private JTextField txtVoto;
19     private JLabel labelTotalVotoBranco;
20     private JLabel labelTotalVotoCandidato1;
21     private JLabel labelTotalVotoCandidato2;
22     private JLabel labelTotalVotacao;
23
24     private JLabel lblNome;
25     private JLabel lblNome1;
26     private JLabel lblNome2;
27     private JLabel lblNome3;
28     private JLabel lblMensagemVotacao;
29
30     private JButton btnApurar;
31     private JButton btnVotar;
32
33     private Container ctn;
34
35     public Votacao objetoVotacao;
```



```
InterfacePrincipal()
```

```
{
```

```
    setSize(300,430);
```

```
    setTitle("URNA ELETRONICA");
```

```
    ctn = getContentPane();
```

```
    ctn.setLayout(null);
```

```
    lblCandidato = new JLabel ("Candidatos");
```

```
    lblCandidato.setBounds(10,10,100,25);
```

```
    lblNome1 = new JLabel ("1 - Joaquim Santos");
```

```
    lblNome1.setBounds(150,10,150,25);
```

```
    lblNome2 = new JLabel ("2 - Karina Constas");
```

```
    lblNome2.setBounds(150,30,150,25);
```

```
    lblNome3 = new JLabel ("3 - Voto em Branco");
```

```
    lblNome3.setBounds(150,50,150,25);
```

```
    lblNome = new JLabel ("Voto:");
```

```
    lblNome.setBounds(10,80,100,25);
```

```
    lblMensagemVotacao = new JLabel ("Digite o seu voto!!!");
```

```
    lblMensagemVotacao.setBounds(60,180,200,25);
```

```
    txtVoto = new JTextField();
```

```
    txtVoto.setBounds(40,80,25,25);
```

```
    btnVotar = new JButton("Votar");
```

```
    btnVotar.setBounds(170,80,75,25);
```

```
    btnApurar = new JButton("Apuração dos votos");
```

```
    btnApurar.setBounds(10,260,250,25);
```

```
lblTotal = new JLabel ("Total de votos");  
lblTotal.setBounds(10,290,250,25);  
labelTotalVotacao = new JLabel("--");  
labelTotalVotacao.setBounds(170,290,30,25);
```

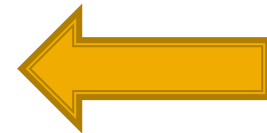
```
lblTotalBranco = new JLabel ("Total em branco");  
lblTotalBranco.setBounds(10,310,350,25);  
labelTotalVotoBranco = new JLabel("--");  
labelTotalVotoBranco.setBounds(170,310,30,25);
```

```
lblCandidato1 = new JLabel ("Votos em candidato 1");  
lblCandidato1.setBounds(10,330,250,25);  
labelTotalVotoCandidato1 = new JLabel("--");  
labelTotalVotoCandidato1.setBounds(170,330,30,25);
```

```
lblCandidato2 = new JLabel ("Votos em candidato 2");  
labelTotalVotoCandidato2 = new JLabel("--");  
labelTotalVotoCandidato2.setBounds(170,350,30,25);  
lblCandidato2.setBounds(10,350,250,25);
```

```
ctn.add(btnVotar);  
ctn.add(btnApurar);  
ctn.add(lblCandidato);  
ctn.add(txtVoto);  
ctn.add(lblNome);  
ctn.add(lblNome1);  
ctn.add(lblNome2);  
ctn.add(lblNome3);  
ctn.add(lblTotal);  
ctn.add(lblTotalBranco);  
ctn.add(lblCandidato1);  
ctn.add(lblCandidato2);  
ctn.add(labelTotalVotacao);  
ctn.add(labelTotalVotoCandidato1);  
ctn.add(labelTotalVotoCandidato2);  
ctn.add(labelTotalVotoBranco);  
ctn.add(lblMensagemVotacao);
```

```
btnVotar.addActionListener (this);  
btnApurar.addActionListener ( this);
```



```
objetoVotacao = new Votacao();
```



```
setVisible(true);
```



```
}
```

```
-
```

```
.
@Override
public void actionPerformed(ActionEvent e)
{

    if(e.getActionCommand().equals("Votar"))
    {
        int numero = Integer.parseInt(txtVoto.getText());

        int resultado = objetoVotacao.Votar(numero);

        if(resultado == 1)
            lblMensagemVotacao.setText("Voto computado com sucesso!!!!");
        else lblMensagemVotacao.setText("Erro! Candidato inválido!!");

        txtVoto.setText("");
    }

    if(e.getActionCommand().equals("Apuração dos votos"))
    {

        labelTotalVotacao.setText(Integer.toString(objetoVotacao.getTotalVotos()));
        labelTotalVotoBranco.setText(Integer.toString(objetoVotacao.getVotoBranco()));
        labelTotalVotoCandidato1.setText(Integer.toString(objetoVotacao.getCandidato1()));
        labelTotalVotoCandidato2.setText(Integer.toString(objetoVotacao.getCandidato2()));
    }

}

public static void main(String[] args) {

    InterfacePrincipal telaInicial = new InterfacePrincipal();
}
```



Exercícios 😊

Exercício 1

PROVA URNA ELETR...

Candidatos

- 1 - Joaquim Santos
- 2 - Karina Constras
- 3 - Voto em Branco

Voto:

Votar

Apuracao dos votos

Total de votos --

Total em branco --

Votos em candidato 1 --

Votos em candidato 2 --



URNA ELETRONICA

Candidatos

- 1 - Joaquim Santos
- 2 - Karina Constras
- 3 - Voto em Branco

Voto:

Votar

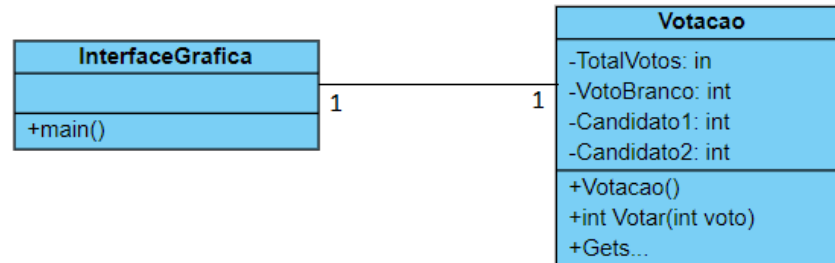
Apuracao dos votos

Total de votos 3

Total em branco 0

Votos em candidato 1 1

Votos em candidato 2 2



Exercício 2

Implemente uma interface gráfica que cadastre o **peso** e a **altura** de várias pessoas.

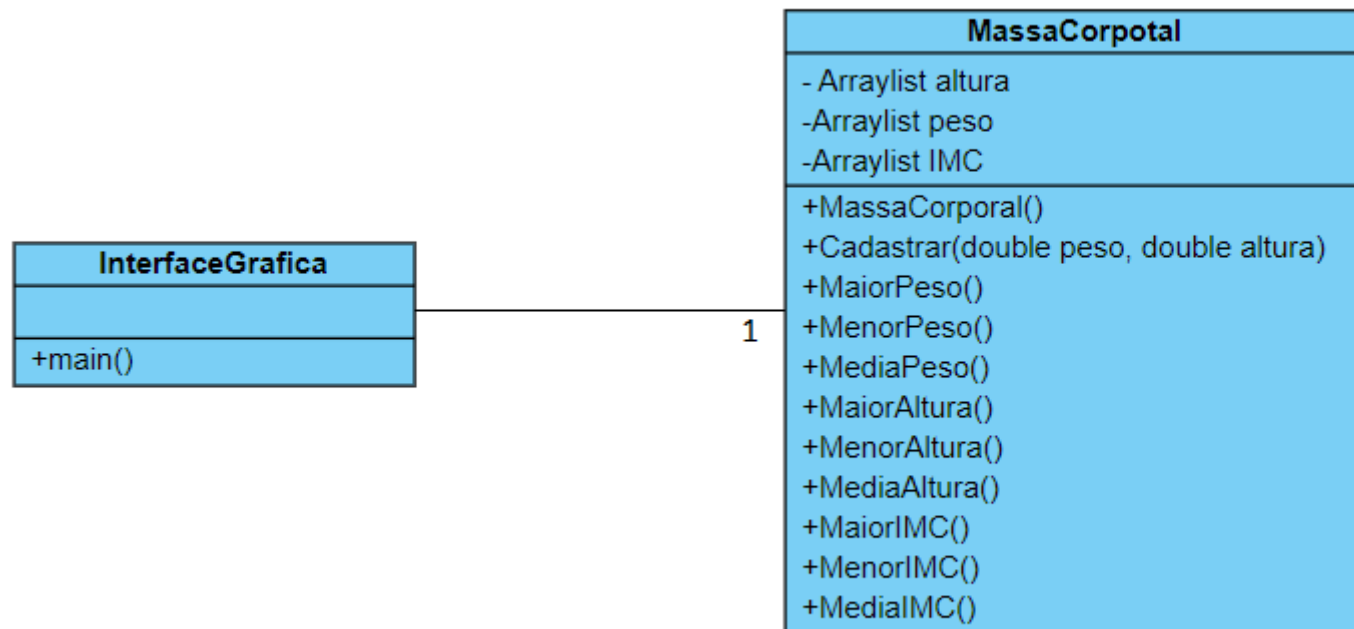
*Ao clicar no botão “**Cadastrar**”, o programa deverá exibir na tela:*

- O Índice de Massa Corporal (IMC) $IMC = peso / (altura * altura)$.
- Mensagem:
 - “Peso Ideal”, caso o IMC esteja entre 18.5 e 25;
 - “Você está acima do peso ideal!”, caso o IMC esteja acima de 25;
 - “Você está abaixo do peso ideal”, caso o IMC esteja abaixo de 18.5.

*Ao clicar no botão “**Relatório**”, o programa deverá exibir na tela:*

- O maior peso cadastrado;
- O menor peso cadastrado;
- A média de todos os pesos cadastrados;
- A maior altura cadastrada;
- A menor altura cadastrada;
- A média de todas as alturas cadastradas;
- A média de todos os IMC cadastrados;
- O maior IMC cadastrado;
- O menor IMC cadastrado.

Dica...



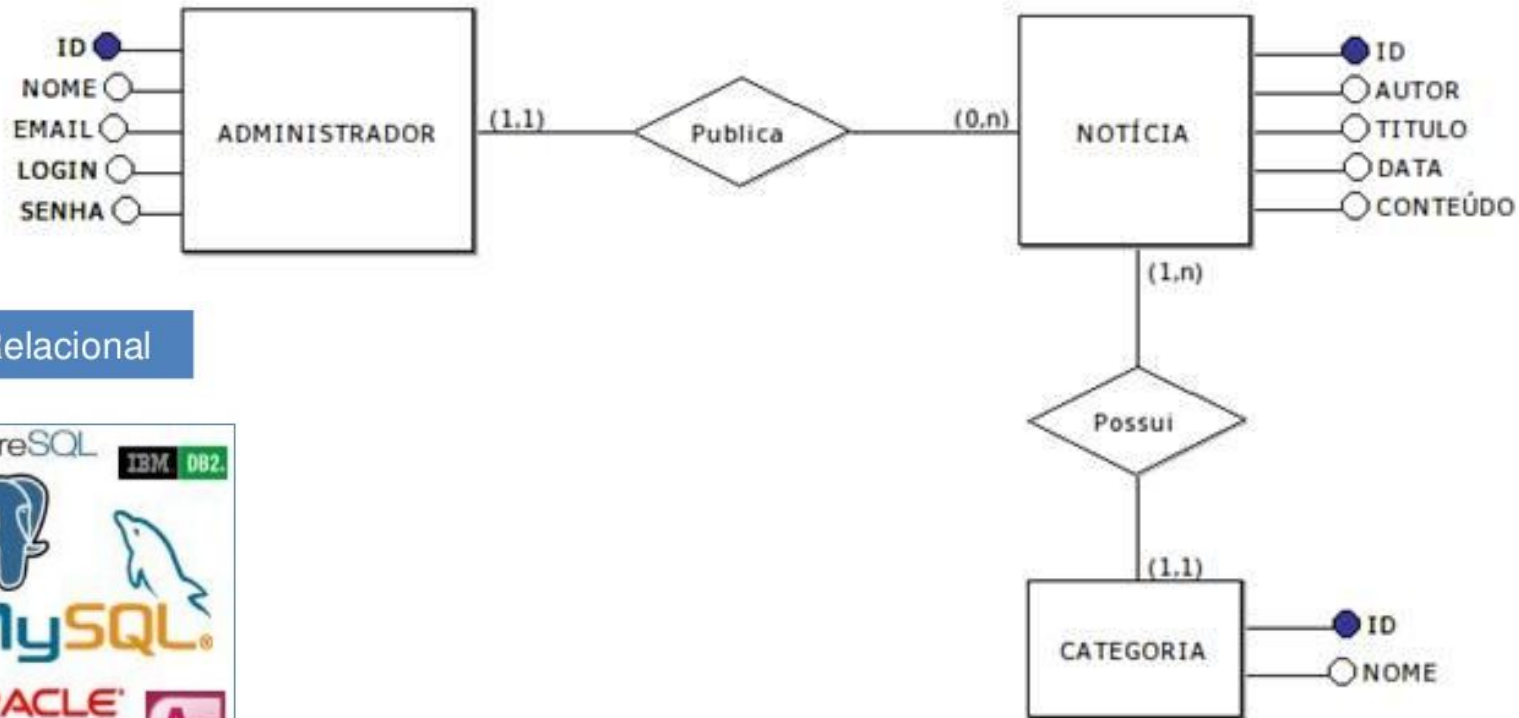
Dica...

Relembrando o uso de ArrayList

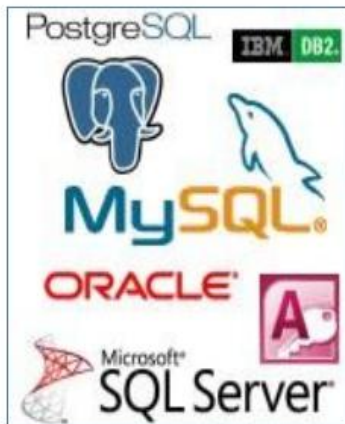
```
Scanner entrada = new Scanner(System.in);  
ArrayList<Integer> lista = new ArrayList<Integer>(); //Cria um arraylist de INTEIROS  
int numero;  
int indice;  
while (true)  
{  
    //insere mais um elemento na lista até que -1 seja digitado  
    System.out.print("Insira um número inteiro positivo para ser incluído ou -1 para terminar: ");  
    numero = entrada.nextInt();  
    if (numero == -1)  
    {  
        break;  
    }  
    lista.add(numero); //adicionar um número na lista  
}  
System.out.println("\nA lista que você criou contém os seguintes elementos: ");  
for (int i: lista)  
{  
    System.out.print(i + " ");  
}
```

Banco de Dados

Em 1970 foram criadas as primeiras versões dos **SGDB relacional** (tabelas e suas relações com chaves) e da linguagem de **consulta estruturada SQL** (*Structured Query Language*).



Relacional



O diagrama apresenta as seguintes tabelas e seus atributos:

- TIPOS_ENUMERADOS**: ID (chave primária), TIPO, VALOR.
- RECURSO**: ID (chave primária), DATA, ALEGACAO, ESTADO_RECURSO, MULTA_ID.
- MULTA**: ID (chave primária), TIPO, NUMERO, DATA, VENCIMENTO, ESTADO_MULTA, TIPO_INFRACAO_ID, VEICULO_ID, PESSOA_ID, EQUIPAMENTO_ID, NOME_AGENTE, ORGAO.
- EQUIPAMENTO**: ID (chave primária), NUMERO, LOCALIZACAO.
- TIPO_INFRACAO**: ID (chave primária), CODIGO, DESCRICAO.
- VEICULO**: ID (chave primária), RENAVAM, PLACA, CHASSI, MODELO, CILINDRADAS, PROPRIETARIO_ID.
- RECURSO_TRANSFERENCIA**: ID (chave primária), DATA, ALEGACAO, ESTADO_RECURSO, MULTA_ID, DATA_VENDA, INFRATOR_ID.
- PESSOA_FISICA**: ID (chave primária), CPF, DATA_NASCIMENTO, ESTADO_CIVIL.
- PESSOA_JURIDICA**: ID (chave primária), CGC.
- CNH**: ID (chave primária), REGISTRO.
- RENOVACOES_CNH**: CNH_ID, DATA.

As relações entre as tabelas são indicadas por linhas com chaves de relacionamento:

- TIPOS_ENUMERADOS** relaciona-se com **RECURSO**, **MULTA** e **PESSOA_FISICA**.
- RECURSO** relaciona-se com **MULTA** e **RECURSO_TRANSFERENCIA**.
- MULTA** relaciona-se com **EQUIPAMENTO**, **TIPO_INFRACAO**, **VEICULO**, **PESSOA** e **RECURSO_TRANSFERENCIA**.
- RECURSO_TRANSFERENCIA** relaciona-se com **PESSOA_FISICA**.
- PESSOA_FISICA** relaciona-se com **PESSOA**.
- PESSOA** relaciona-se com **PESSOA_JURIDICA**, **CNH** e **RENOVACOES_CNH**.

Banco de Dados

SQLQuery2.sql - MA...te (MAC\user (55))* x SQLQuery1.sql - MA...ro (MAC\user (52))*

```
use Teste
Insert Into Funcionarios values ('Jose Carlos','Macoratti',5000,'M','Brasilia')
select * from funcionarios
```

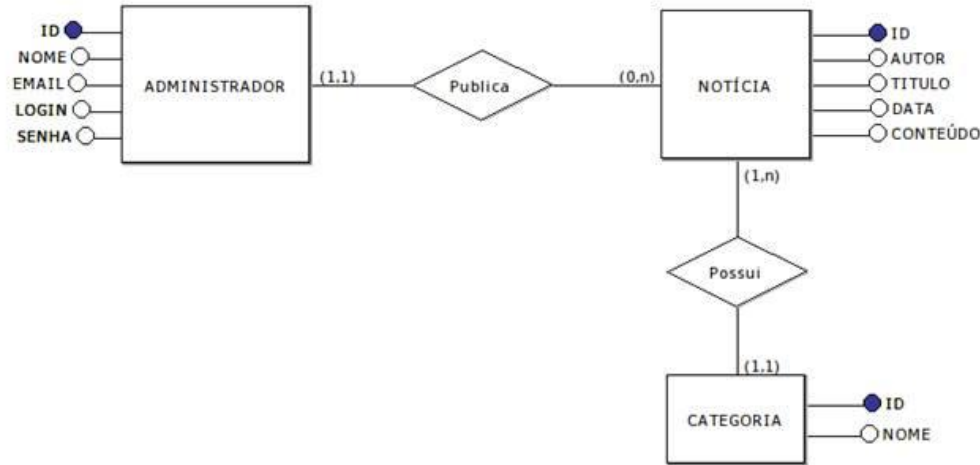
100 %

Results Messages

	Id	Nome	Sobrenome	Salario	Sexo	Cidade
1	1	Jose Carlos	Macoratti	5000	M	Brasilia

Query executed successfully. | MAC\SQLEXPRESS (9.0 SP2) | MAC\user (55) | Teste | 00:00:00 | 1 rows

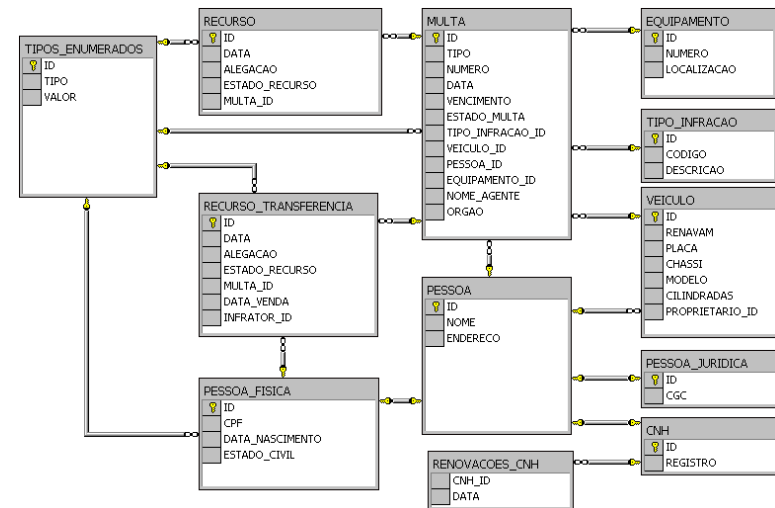
Banco de Dados Relacionais



ORACLE®



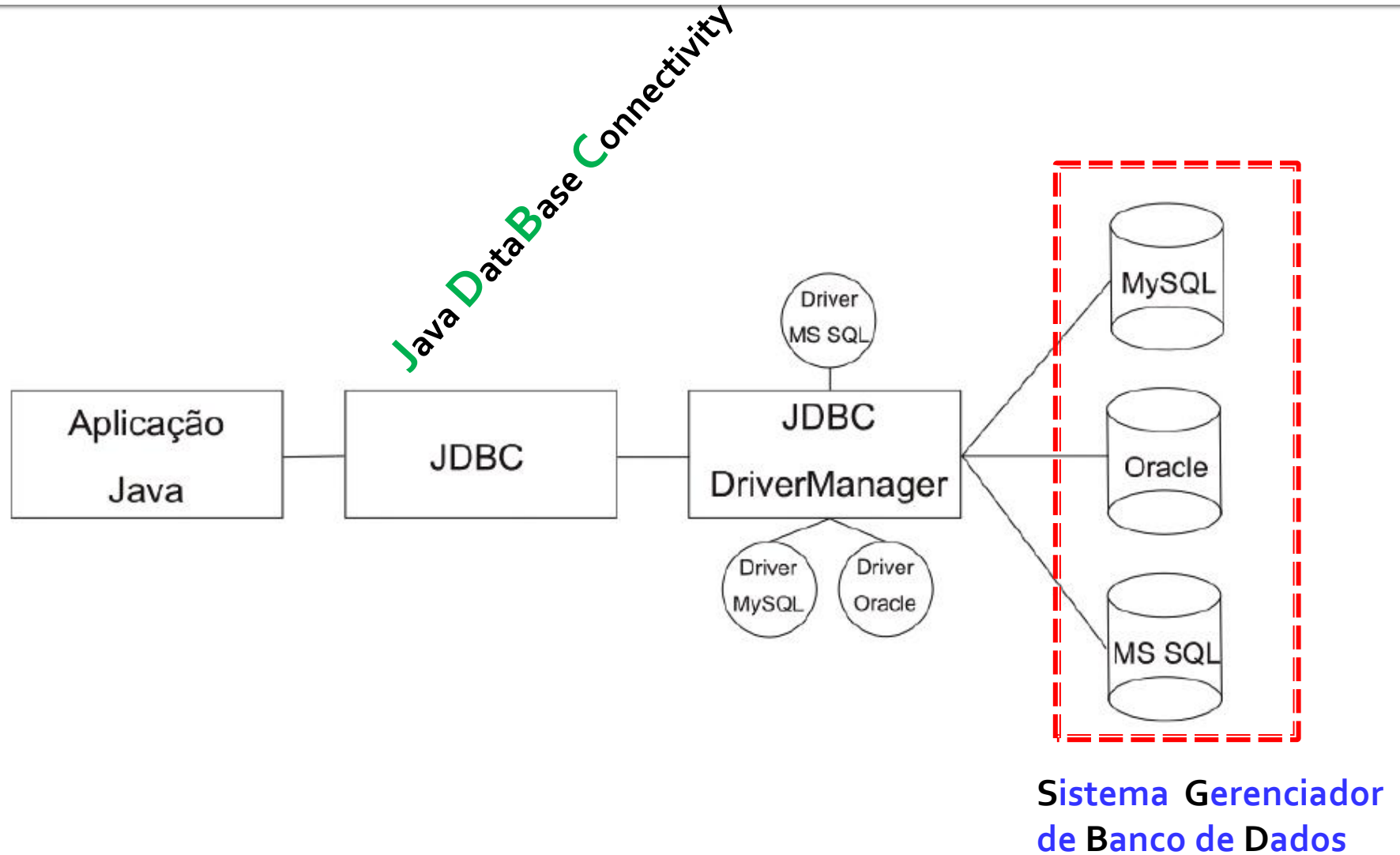
PostgreSQL



Interface Gráfica-Eventos-Banco de Dados

- Recuperar os dados da interface gráfica utilizando o tratamento de **eventos** em variáveis locais.
- Criar uma **classe que faça a conexão com o sistema de banco de dados**.
- Produzir uma classe que **execute os comandos no banco de dados** com a conexão criado no item anterior.
- Na classe que executa os comandos, **criar métodos para receber os dados** que serão inseridos na interface gráfica e **execute os comandos no sistema de banco de dados**

Java-Banco de Dados



Java-Banco de Dados

1º passo: Fazer o download e instalar o SGBD no seu computador

2º passo: Criar o Banco de Dados no SGBD

3º passo: Fazer o download do JDBC

4º passo: Inserir o JDBC no Eclipse

4º passo: Desenvolver o programa Java no Eclipse para conectar e realizar operações (inserir, atualizar, remover, consultar) no Banco de Dados que foi criado no passo 2.

Conexão com Banco de Dados

Fazer o Download do **MySql** e Conexões Java no site:

<https://www.mysql.com/products/connector/>

- MySQL HeatWave

- MySQL Enterprise Edition

- Datasheet (PDF)
- Technical Specification
- MySQL Database
- MySQL Document Store
- Oracle Enterprise Manager
- Enterprise Monitor
- Enterprise Backup
- Enterprise HA
- Enterprise Scalability
- Enterprise Authentication
- Enterprise TDE
- Enterprise Encryption
- Enterprise Masking
- Enterprise Firewall

MySQL Connectors

MySQL provides standards-based drivers for JDBC, ODBC, and .Net enabling developers to build database a choice. In addition, a native C library allows developers to embed MySQL directly into their applications.

Developed by MySQL

ADO.NET Driver for MySQL (Connector/NET)

[Download](#)

ODBC Driver for MySQL (Connector/ODBC)

[Download](#)

JDBC Driver for MySQL (Connector/J)

[Download](#)

Node.js Driver for MySQL (Connector/Node.js)

[Download](#)

Download MySQL Connector/J

Python Driver for MySQL (Connector/Python)

[Download](#)

C++ Driver for MySQL (Connector/C++)

[Download](#)

C Driver for MySQL (Connector/C)

[Download](#)

C API for MySQL (mysqlclient)

[Download](#)

These drivers are developed and maintained by the MySQL Community.

General Availability (GA) Releases

Archives



Connector/J 8.0.28

Select Operating System:

Microsoft Windows



Recommended Windows Download:

MySQL Installer for Windows

**All MySQL Products. For All Windows Platforms.
In One Package.**

Starting with MySQL 5.6 the MySQL Installer package replaces the standalone MSI packages.



Windows (x86, 32 & 64-bit), MySQL Installer MSI

[Go to Download Page >](#)



General Availability (GA) Releases

Archives



MySQL Installer 8.0.28

Select Operating System:

Microsoft Windows



[Looking for previous GA versions?](#)

Windows (x86, 32-bit), MSI Installer

8.0.28

2.3M

[Download](#)

(mysql-installer-web-community-8.0.28.0.msi)

MD5: 514567a7503999d271a20b86057f15d0 | [Signature](#)

Windows (x86, 32-bit), MSI Installer

8.0.28

435.7M

[Download](#)

(mysql-installer-community-8.0.28.0.msi)

MD5: ef223cbd7873a057d5aa0abcf596201 | [Signature](#)



We suggest that you use the [MD5 checksums and GnuPG signatures](#) to verify the integrity of the packages you download.

MySQL Community Downloads

Login Now or Sign Up for a free account.

An Oracle Web Account provides you with the following advantages:

- Fast access to MySQL software downloads
- Download technical White Papers and Presentations
- Post messages in the MySQL Discussion Forums
- Report and track bugs in the MySQL bug system

Login »

using my Oracle Web account

Sign Up »

for an Oracle Web account

MySQL.com is using Oracle SSO for authentication. If you already have an Oracle Web account, click the Login link. Otherwise, you can sign up for a free account by clicking the Sign Up link and following the instructions.

[No thanks, just start my download.](#)



MySQL. Installer 1.6

Finding all installed packages.

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Other names may be trademarks of their respective owners.

ORACLE®

MySQL Installer

MySQL. Installer Adding Community

Choosing a Setup Type

Download

Installation

Installation Complete

Choosing a Setup Type

Please select the Setup Type that suits your use case.

☒ **Developer Default**

Installs all products needed for MySQL development purposes.

☐ **Server only**

Installs only the MySQL Server product.

☐ **Client only**

Installs only the MySQL Client products, without a server.

☐ **Full**

Installs all included MySQL products and features.

☐ **Custom**

Manually select the products that should be installed on the system.

Setup Type Description

The GUI application to develop for and manage the server.

* MySQL for Visual Studio

To work with the MySQL Server from VS.

* MySQL Connectors

Connector/Net, Java, C/C++, ODBC and others.

* Examples and tutorials

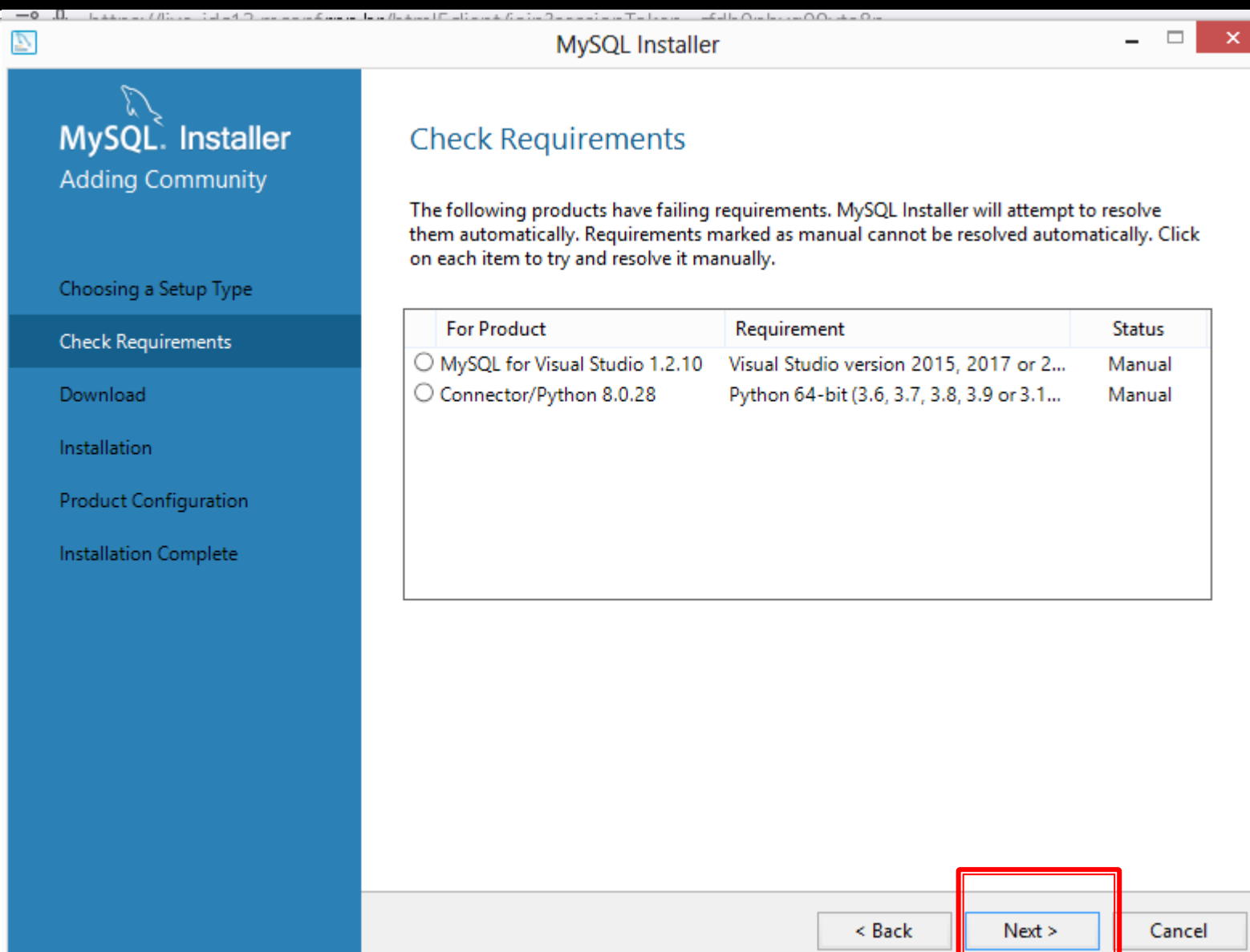
To help you get started with your development.

* Documentation

Allows you to read the documentation offline.

Next >

Cancel



MySQL Installer

Adding Community

Choosing a Setup Type

Check Requirements

Download

Installation

Product Conf

Installation C

Check Requirements

The following products have failing requirements. MySQL Installer will attempt to resolve them automatically. Requirements marked as manual cannot be resolved automatically. Click on each item to try and resolve it manually.

For Product	Requirement	Status
MySQL 5.6.16-1.0.10	MySQL 5.6.16-1.0.10	Manual
MySQL 5.6.16-1.0.10	MySQL 5.6.16-1.0.10	Manual

MySQL Installer



One or more product requirements have not been satisfied

Those products with missing requirements will not be installed or upgraded.

Do you want to continue?

Yes

No

< Back

Next >

Cancel

MySQL Installer

MySQL. Installer Adding Community

Choosing a Setup Type

Download

Installation

Product Configuration

Installation Complete

Download

The following products will be downloaded.

Product	Status	Progress	Notes
 MySQL Server 8.0.28	Ready to download		
 MySQL Workbench 8.0.28	Ready to download		
 MySQL Shell 8.0.28	Ready to download		
 MySQL Router 8.0.28	Ready to download		
 Connector/ODBC 8.0.28	Ready to download		
 Connector/C++ 8.0.28	Ready to download		
 Connector/J 8.0.28	Ready to download		
 Connector/NET 8.0.28	Ready to download		
 MySQL Documentation 8.0.28	Ready to download		
 Samples and Examples 8.0.28	Ready to download		

Click [Execute] to install the following packages.

< Back

Execute

Cancel



MySQL Installer

MySQL. Installer

Adding Community

Choosing a Setup Type

Download

Installation

Product Configuration

Installation Complete

Download

The following products will be downloaded.

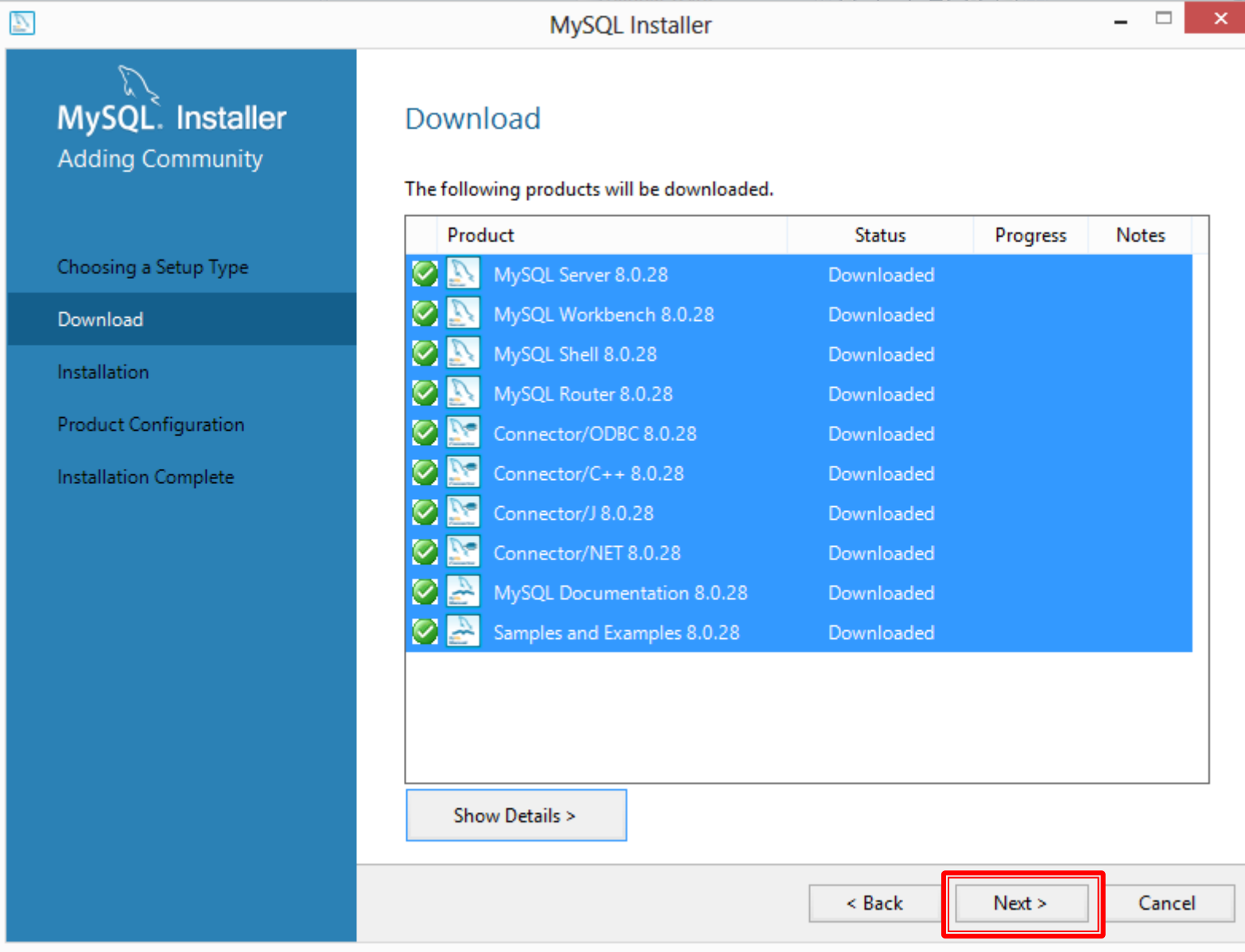
	Product	Status	Progress	Notes
	MySQL Server 8.0.28	Downloading...	4%	
	MySQL Workbench 8.0.28	Downloading...	15%	
	MySQL Shell 8.0.28	Downloading...	18%	
	MySQL Router 8.0.28	Downloading...	26%	
	Connector/ODBC 8.0.28	Downloading...	31%	
	Connector/C++ 8.0.28	Downloading...	20%	
	Connector/J 8.0.28	Downloaded		
	Connector/NET 8.0.28	Downloading...	55%	
	MySQL Documentation 8.0.28	Downloading...	25%	
	Samples and Examples 8.0.28	Downloaded		

Show Details >

< Back

Execute

Cancel





MySQL Installer

MySQL. Installer

Adding Community

Choosing a Setup Type

Installation

Product Configuration

Installation Complete

Installation

The following products will be installed.

Product	Status	Progress	Notes
 MySQL Server 8.0.28	Ready to Install		
 MySQL Workbench 8.0.28	Ready to Install		
 MySQL Shell 8.0.28	Ready to Install		
 MySQL Router 8.0.28	Ready to Install		
 Connector/ODBC 8.0.28	Ready to Install		
 Connector/C++ 8.0.28	Ready to Install		
 Connector/J 8.0.28	Ready to Install		
 Connector/NET 8.0.28	Ready to Install		
 MySQL Documentation 8.0.28	Ready to Install		
 Samples and Examples 8.0.28	Ready to Install		

Click [Execute] to install the following packages.

< Back

Execute

Cancel



MySQL Installer

MySQL. Installer

Adding Community

Choosing a Setup Type

Installation

Product Configuration

Installation Complete

Installation

The following products will be installed.

	Product	Status	Progress	Notes
✓	 MySQL Server 8.0.28	Complete		
	 MySQL Workbench 8.0.28	Installing	99%	
	 MySQL Shell 8.0.28	Ready to Install		
	 MySQL Router 8.0.28	Ready to Install		
	 Connector/ODBC 8.0.28	Ready to Install		
	 Connector/C++ 8.0.28	Ready to Install		
	 Connector/J 8.0.28	Ready to Install		
	 Connector/NET 8.0.28	Ready to Install		
	 MySQL Documentation 8.0.28	Ready to Install		
	 Samples and Examples 8.0.28	Ready to Install		

Show Details >

< Back

Execute

Cancel



MySQL Installer

MySQL. Installer

Adding Community

Choosing a Setup Type

Installation

Product Configuration

Installation Complete

Installation

The following products will be installed.

	Product	Status	Progress	Notes
✓	 MySQL Server 8.0.28	Complete		
✓	 MySQL Workbench 8.0.28	Complete		
✓	 MySQL Shell 8.0.28	Complete		
✓	 MySQL Router 8.0.28	Complete		
✓	 Connector/ODBC 8.0.28	Complete		
✓	 Connector/C++ 8.0.28	Complete		
✓	 Connector/J 8.0.28	Complete		
✓	 Connector/NET 8.0.28	Complete		
✓	 MySQL Documentation 8.0.28	Complete		
✓	 Samples and Examples 8.0.28	Complete		

Show Details >

< Back

Next >

Cancel

MySQL Installer

Product Configuration

We'll now walk through a configuration wizard for each of the following products.

You can cancel at any point if you wish to leave this wizard without configuring all the products.

Product	Status
MySQL Server 8.0.28	Ready to configure
MySQL Router 8.0.28	Ready to configure
Samples and Examples 8.0.28	Ready to configure

Next >

Cancel



MySQL. Installer

MySQL Server 8.0.28

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Apply Configuration

Type and Networking

Server Configuration Type

Choose the correct server configuration type for this MySQL Server installation. This setting will define how much system resources are assigned to the MySQL Server instance.

Config Type: Development Computer

Connectivity

Use the following controls to select how you would like to connect to this server.

☒ TCP/IP

Port: 3306X Protocol Port: 33060

☒ Open Windows Firewall ports for network access

☐ Named Pipe

Pipe Name: MYSQL

☐ Shared Memory

Memory Name: MYSQL

Advanced Configuration

Select the check box below to get additional configuration pages where you can set advanced and logging options for this server instance.

☐ Show Advanced and Logging Options

Next >

Cancel

MySQL Installer

MySQL[®] Installer

MySQL Server 8.0.28

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Apply Configuration

Authentication Method

☒ **Use Strong Password Encryption for Authentication (RECOMMENDED)**

MySQL 8 supports a new authentication based on improved stronger SHA256-based password methods. It is recommended that all new MySQL Server installations use this method going forward.



Attention: This new authentication plugin on the server side requires new versions of connectors and clients which add support for this new 8.0 default authentication (caching_sha2_password authentication).

Currently MySQL 8.0 Connectors and community drivers which use libmysqlclient 8.0 support this new method. If clients and applications cannot be updated to support this new authentication method, the MySQL 8.0 Server can be configured to use the legacy MySQL Authentication Method below.

☐ **Use Legacy Authentication Method (Retain MySQL 5.x Compatibility)**

Using the old MySQL 5.x legacy authentication method should only be considered in the following cases:

- If applications cannot be updated to use MySQL 8 enabled Connectors and drivers.
- For cases where re-compilation of an existing application is not feasible.
- An updated, language specific connector or driver is not yet available.

Security Guidance: When possible, we highly recommend taking needed steps towards upgrading your applications, libraries, and database servers to the new stronger authentication. This new method will significantly improve your security.

< Back

Next >

Cancel

MySQL Installer

MySQL® Installer MySQL Server 8.0.28

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Apply Configuration

Accounts and Roles

Root Account Password

Enter the password for the root account. Please remember to store this password in a secure place.

MySQL Root Password:

•••••

Repeat Password:

•••••

Password strength: **Weak**

MySQL User Accounts

Create MySQL user accounts for your users and applications. Assign a role to the user that consists of a set of privileges.

MySQL User Name	Host	User Role

Add User

Edit User

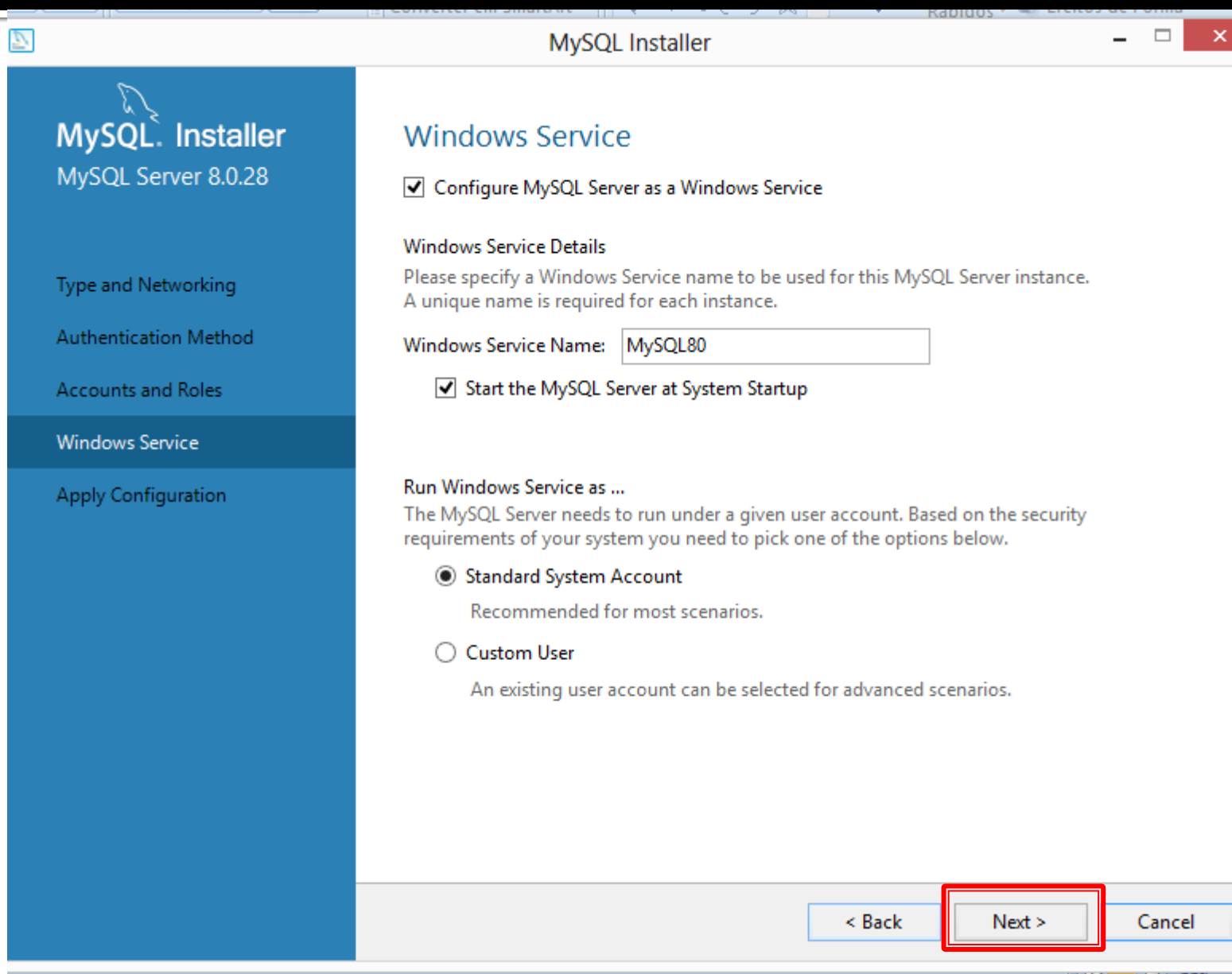
Delete

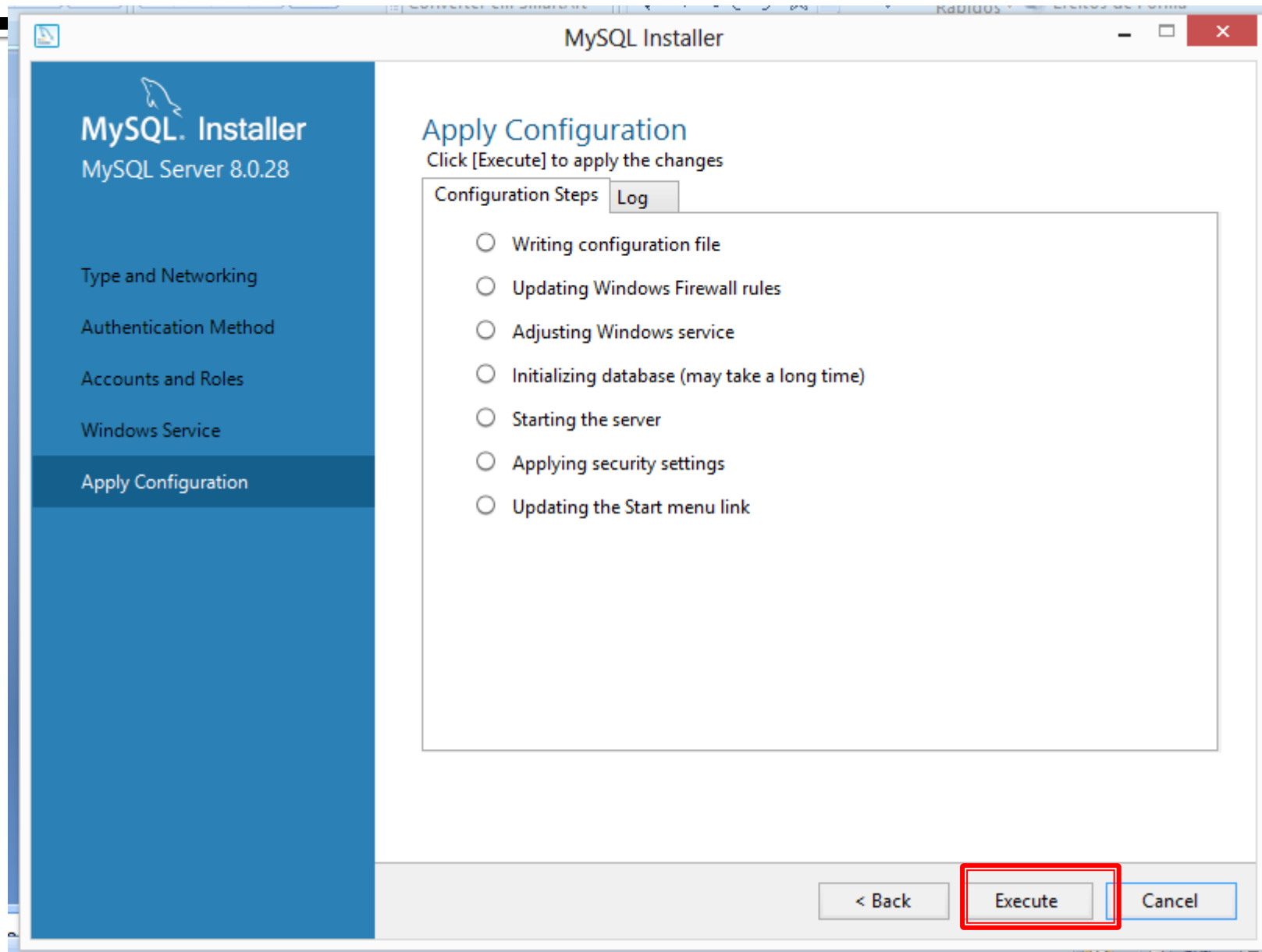
< Back

Next >

Cancel

Senha = admin







MySQL Installer

MySQL Installer

MySQL Server 8.0.28

Type and Networking

Authentication Method

Accounts and Roles

Windows Service

Apply Configuration

Apply Configuration

The configuration operation has completed.

Configuration Steps

Log

- ✓ Writing configuration file
- ✓ Updating Windows Firewall rules
- ✓ Adjusting Windows service
- ✓ Initializing database (may take a long time)
- ✓ Starting the server
- ✓ Applying security settings
- ✓ Updating the Start menu link

The configuration for MySQL Server 8.0.28 was successful.
Click Finish to continue.

Finish



MySQL. Installer

Adding Community

Choosing a Setup Type

Installation

Product Configuration

Installation Complete

Product Configuration

We'll now walk through a configuration wizard for each of the following products.

You can cancel at any point if you wish to leave this wizard without configuring all the products.

Product	Status
MySQL Server 8.0.28	Configuration complete.
MySQL Router 8.0.28	Ready to configure
Samples and Examples 8.0.28	Ready to configure

Next >

Cancel



MySQL Installer

MySQL. Installer MySQL Router 8.0.28

MySQL Router Configuration

MySQL Router Configuration

☐ Bootstrap MySQL Router for use with InnoDB Cluster

This wizard can bootstrap MySQL Router to direct traffic between MySQL applications and InnoDB Cluster. Applications that connect to the router will be automatically directed to an available read/write or read-only member of the cluster.

The bootstrapping process requires a connection to InnoDB Cluster. In order to register the MySQL Router for monitoring, use the current Read/Write instance of the cluster.

Hostname:

Port:

Management User:

Password:

MySQL Router requires specification of a base port (between 80 and 65532). The first port is used for classic read/write connections. The other ports are computed sequentially after the first port. If any port is indicated to be in use, please change the base port.

Classic MySQL protocol connections to InnoDB Cluster:

Read/Write:

Read Only:

X Protocol connections to InnoDB Cluster:

Read/Write:

Read Only:



MySQL® Installer

Samples and Examples

Connect To Server

Apply Configuration

Connect To Server

Select the MySQL server instances from the list to receive sample schemas and data.

	Server	Port	Arch...	Type	Status
☒	MySQL Server 8.0.28	3306	X64	Stand-alone Server	Connection succeeded.

Provide the credentials that should be used (requires root privileges).
Click "Check" to ensure they work.

User name: Credentials provided in Server configuration

Password:

Check

Next >

Cancel



MySQL Installer

MySQL[®] Installer Samples and Examples

Connect To Server

Apply Configuration

Apply Configuration

Click [Execute] to apply the changes

Configuration Steps

Log

- ☐ Checking if there are any features installed that need configuration.
- ☐ Running Scripts

< Back

Execute

Cancel



MySQL Installer

MySQL[®] Installer Samples and Examples

Connect To Server

Apply Configuration

Apply Configuration

The configuration operation has completed.

Configuration Steps [Log](#)

- ✓ Checking if there are any features installed that need configuration.
- ✓ Running Scripts

The configuration for Samples and Examples 8.0.28 was successful.
Click Finish to continue.

Finish



MySQL Installer

MySQL® Installer Adding Community

Choosing a Setup Type

Installation

Product Configuration

Installation Complete

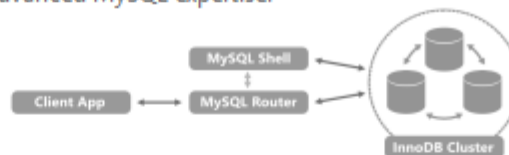
Installation Complete

The installation procedure has been completed.

Copy Log to Clipboard

- ☒ Start MySQL Workbench after setup
- ☒ Start MySQL Shell after setup

The MySQL Shell is an advanced MySQL client application that can be used to work with single MySQL Server instances. Further, it can be used to create and manage InnoDB Cluster, an integrated solution for high availability and scalability of MySQL databases, without requiring advanced MySQL expertise.



Refer to the following links for documentation, tutorials and examples on MySQL Shell:

[MySQL Shell Documentation](#)

[Setting up a Real World Cluster Blog](#)

[The All New MySQL InnoDB ReplicaSet Blog](#)

[Changing Cluster Options Live Blog](#)

Finish

Java-Banco de Dados

1º passo: Fazer o download e instalar o SGBD no seu computador

2º passo: Criar o Banco de Dados no SGBD

3º passo: Fazer o download do JDBC

4º passo: Inserir o JDBC no Eclipse

4º passo: Desenvolver o programa Java no Eclipse para conectar e realizar operações (inserir, atualizar,remover, consultar) no Banco de Dados que foi criado no passo 2.



Welcome to MySQL Workbench

MySQL Workbench is the official graphical user interface (GUI) tool for MySQL. It allows you to design, create and browse your database schemas, work with database objects and insert data as well as design and run SQL queries to work with stored data. You can also migrate schemas and data from other database vendors to your MySQL database.

[Browse Documentation >](#)[Read the Blog >](#)[Discuss on the Forums >](#)

MySQL Connections

Local instance MySQL80

 root

 localhost:3306



Welcome to MySQL Workbench

MySQL Workbench is the official database IDE for the database platform. It allows you to create and browse your database, design and run SQL queries, and manage your database.

[Browse Documentation >](#)

MySQL Connections

Local instance MySQL80

 root

 localhost:3306

Connect to MySQL Server

Please enter password for the following service:



Service: Mysql@localhost:3306

User: root

Password:

☐ Save password in vault

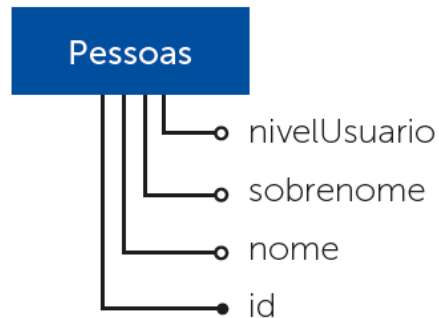
Senha = admin
SE FOR NO SEU COMPUTADOR!

Senha = root
SE FOR NO COMPUTADOR DOS LABORATÓRIOS!

Banco de Dados

Nome do Banco de Dados: **nomebd**

Figura 1.7| Dados para cadastro de uma pessoa



Exemplo da página 50



Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

New Model Ctrl+N

New Query Tab Ctrl+T

Open Model... Ctrl+O

Open SQL Script... Ctrl+Shift+O

Open Recent ▶

Run SQL Script...

Close Connection Tab Ctrl+Shift+F4

Close Tab Ctrl+W

Save Script Ctrl+S

Save Script As... Ctrl+Shift+S

Revert to Saved

Exit Alt+F4



1 x



Limit to 1000 rows

SQLAdditions

**Automatic context help is
current can**

Administration Schemas

Information



Context Help

Snippets

Output



Action Output



#

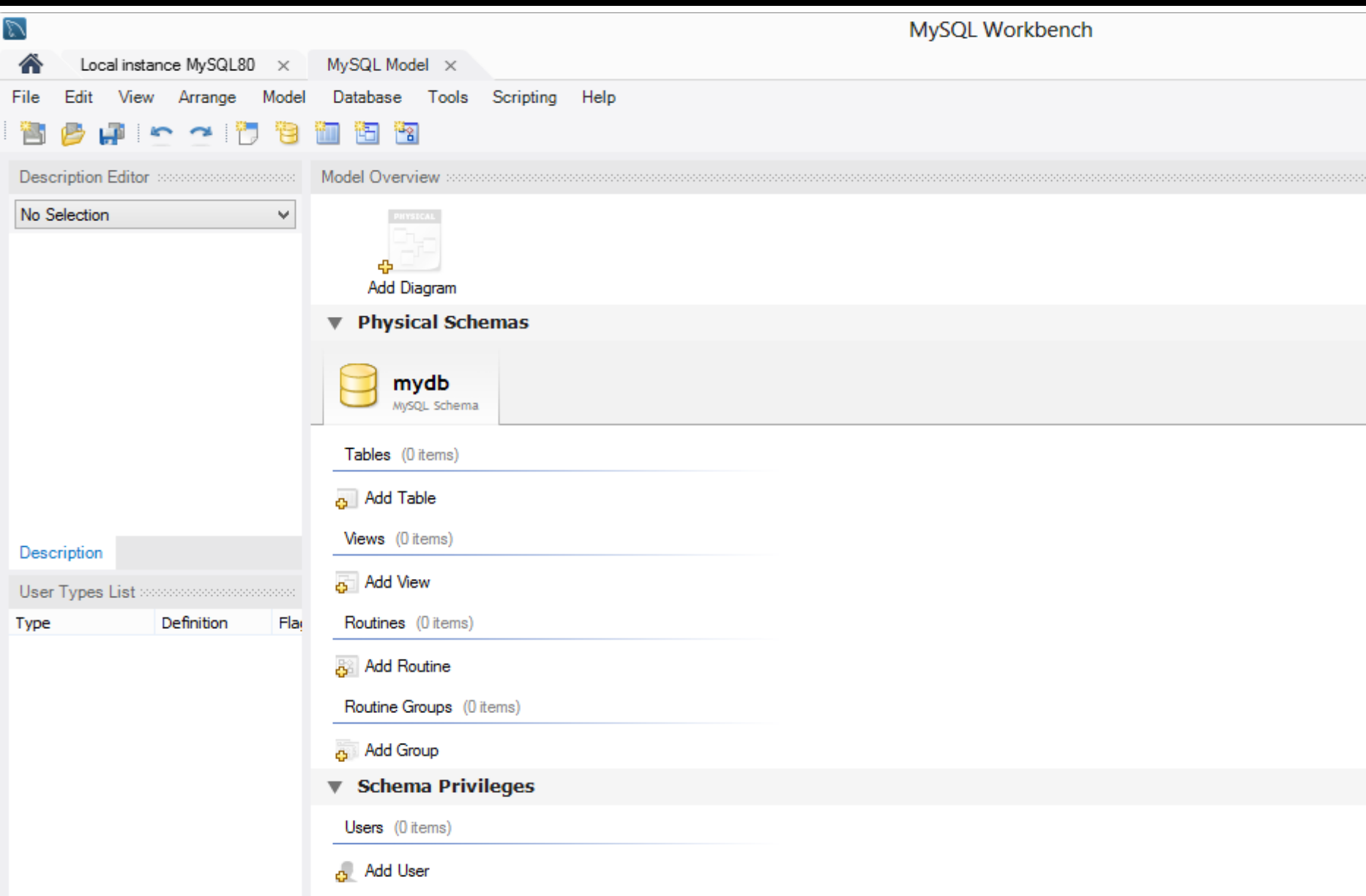
Time

Action

Message

No object selected

Modelar o Banco de Dados



Local instance MySQL80 x MySQL Model x

File Edit View Arrange Model Database Tools Scripting Help

Description Editor

No Selection

Description

User Types List

Type	Definition	Flags
------	------------	-------

Model Overview



Add Diagram

▼ Physical Schemas

mydb
MySQL 5

Tables (0 items)

Add Table

Views (0 items)

Add View

Routines (0 items)

Add Routine

Routine Groups (0 items)

Add Group

▼ Schema Privileges

Users (0 items)

Add User

Cut 'mydb'

Copy 'mydb'

Paste

Edit Schema...

Edit in New Tab...

Copy SQL to Clipboard

Delete 'mydb'

Remove Figure 'mydb'



Description Editor

No Selection

Model Overview



Add Diagram

▼ Physical Schemas

**nomebd**
MySQL Schema

Tables (0 items)

nomebd - Schema x



Name: nomebd

Rename References

Specify the name of the schema here. You can use any combination of

Refactor model, changing all references found in view, triggers, stored proced new one.

Charset/Collation:

utf8

utf8_bin

The character set and its collation selected here will be used when no of

Comments:

Description

User Types List

Type

Definition

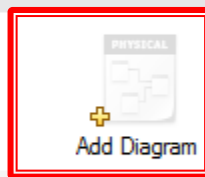
Flag



Description Editor

No Selection

Model Overview



Add Diagram

▼ Physical Schemas

**nomebd**
MySQL Schema

Tables (0 items)

nomebd - Schema x



Name:

nomebd

Rename References

Specify the name of the schema here. You can use any combination of

Refactor model, changing all references found in view, triggers, stored proced new one.

Charset/Collation:

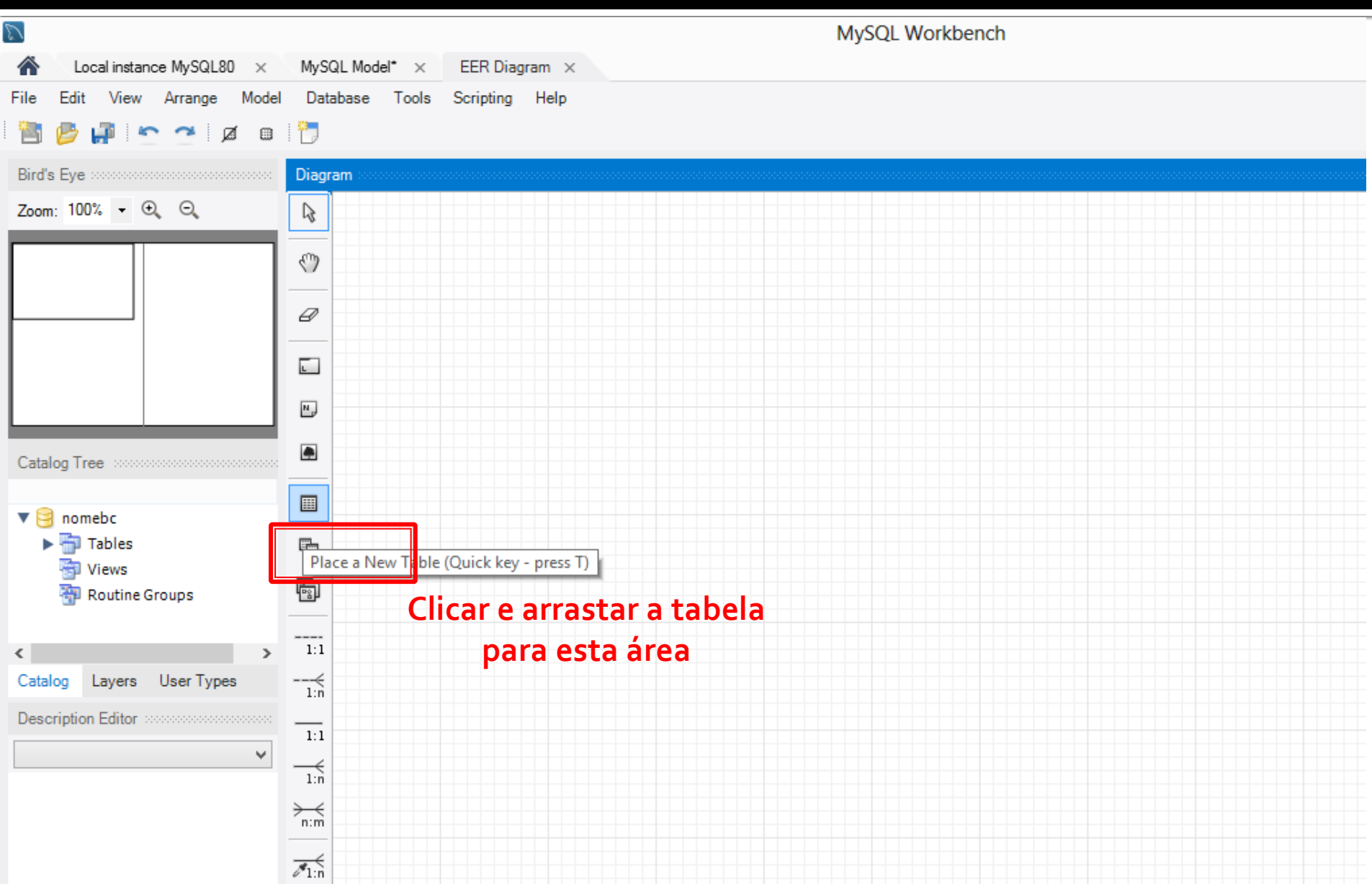
utf8

utf8_bin

The character set and its collation selected here will be used when no of

Comments:

Criar a Tabela Pessoa



Local instance MySQL80 x MySQL Model* x EER Diagram x

File Edit View Arrange Model Database Tools Scripting Help

Bird's Eye

Zoom: 100%

Catalog Tree

- nomebd
 - Tables
 - Views
 - Routine Groups

Diagram

pessoa - Table x

Table Name: pessoa Schema: nomebd

Column Name	Datatype	PK	NN	UO	B	UN	ZF	AT	G
id	INT	☒	☒	☐	☐	☐	☐	☒	
nome	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	
sobrenome	VARCHAR(45)	☐	☐	☐	☐	☐	☐	☐	
nivelusuario	INT	☐	☐	☐	☐	☐	☐	☐	

Column Name: id

Charset/Collation: Default Charset Default Collation

Comments:

Data Type:

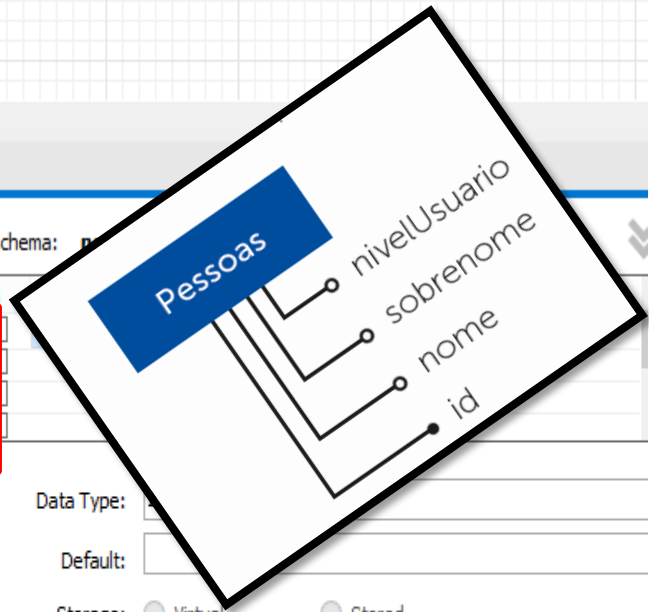
Default:

Storage: ☐ Virtual ☐ Stored

☒ Primary Key ☒ Not Null ☐ Unique

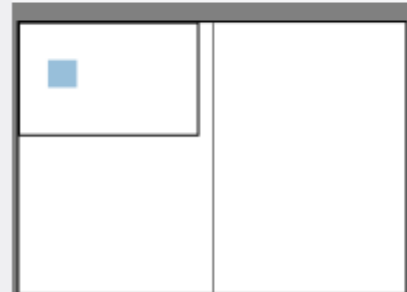
☐ Binary ☐ Unsigned ☐ Zero Fill

☒ Auto Increment ☐ Generated





Bird's Eye

Zoom: 100%  

Catalog Tree

nomebc

- Tables
- Views
- Routine Groups

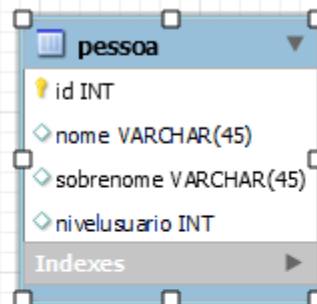
< >

Catalog Layers User Types

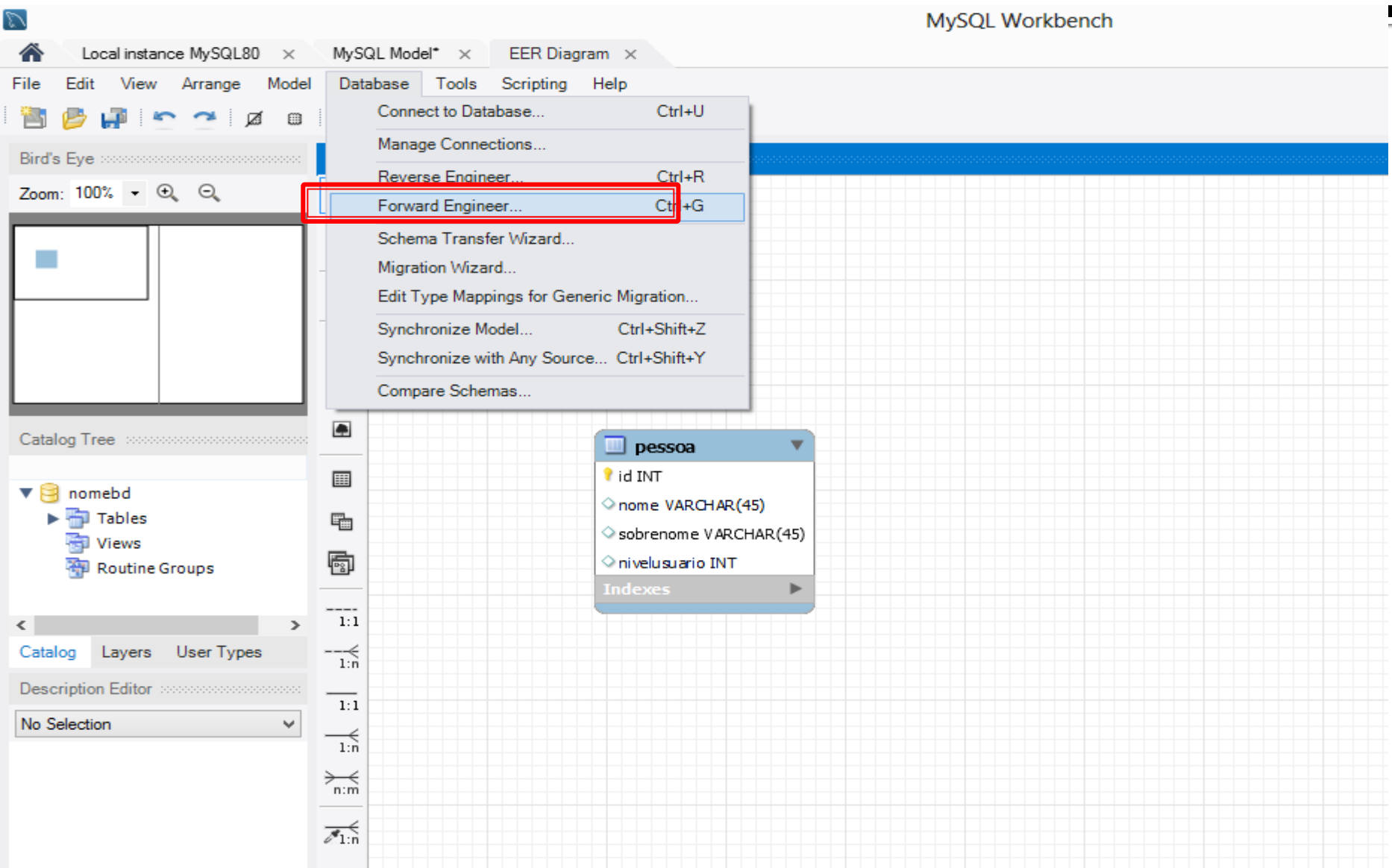
Description Editor

table2: Table

Diagram



Criar o Banco de Dados



Connection Options

Options

Select Objects

Review SQL Script

Commit Progress

Set Parameters for Connecting to a DBMS

Stored Connection: Local instance MySQL80

Select from saved connection settings

Connection Method: Standard (TCP/IP)

Method to use to connect to the RDBMS

Parameters

SSL

Advanced

Hostname: localhost

Port: 3306

Name or IP address of the server host - and TCP/IP port.

Username: root

Name of the user to connect with.

Password:

Store in Vault ...

Clear

The user's password. Will be requested later if it's not set.

Default Schema:

The schema to use as default schema. Leave blank to select it later.

Back

Next

Cancel

Connection Options

Options

Select Objects

Review SQL Script

Commit Progress

Set Options for Database to be Created

Tables

- ☐ Skip creation of FOREIGN KEYS
- ☐ Skip creation of FK Indexes as well
- ☐ Generate separate CREATE INDEX statements
- ☐ Generate INSERT statements for tables
- ☐ Disable FK checks for INSERTs

Other Objects

- ☐ Don't create view placeholder tables
- ☐ Do not create users. Only create privileges (GRANTS)

Code Generation

- ☐ DROP objects before each CREATE object
- ☐ Generate DROP SCHEMA
- ☐ Omit schema qualifier in object names
- ☐ Generate USE statements
- ☐ Add SHOW WARNINGS after every DDL statement
- ☒ Include model attached scripts

Back

Next

Cancel

Connection Options

Options

Select Objects

Review SQL Script

Commit Progress

Select Objects to Forward Engineer

To exclude objects of a specific type from the SQL Export, disable the corresponding checkbox. Press Show Filter and add objects or patterns to the ignore list to exclude them from the export.



☒ Export MySQL Table Objects

2 Total Objects, 2 Selected

Show Filter



☐ Export MySQL User Objects

0 Total Objects, 0 Selected

Show Filter



☐ Export MySQL Database Objects

0 Total Objects, 0 Selected

Show Filter



☐ Export MySQL Schema Objects

0 Total Objects, 0 Selected

Show Filter

☐ Export User Objects

0 Total Objects, 0 Selected

Show Filter

Connect to MySQL Server

Please enter password for the following service:

Service: Mysql@localhost:3306

User: root

Password: *****

☐ Save password in vault

OK

Cancel

Senha = admin
SE FOR NO SEU COMPUTADOR!

Senha = root!

Back

Next

Cancel

Connection Options

Options

Select Objects

Review SQL Script

Commit Progress

Select Objects to Forward Engineer

To exclude objects of a specific type from the SQL Export, disable the corresponding checkbox. Press Show Filter and add objects or patterns to the ignore list to exclude them from the export.

☒ Export MySQL Table Objects

2 Total Objects, 2 Selected

Show Filter

☐ Export MySQL View Objects

0 Total Objects, 0 Selected

Show Filter

☐ Export MySQL Routine Objects

0 Total Objects, 0 Selected

Show Filter

☐ Export MySQL Trigger Objects

0 Total Objects, 0 Selected

Show Filter

☐ Export User Objects

0 Total Objects, 0 Selected

Show Filter

Back

Next

Cancel

Connection Options

Options

Select Objects

Review SQL Script

Commit Progress

Review the SQL Script to be Executed

This script will now be executed on the DB server to create your databases.
You may make changes before executing.

```
1      -- MySQL Workbench Forward Engineering
2
3      SET @OLD_UNIQUE_CHECKS=@@UNIQUE_CHECKS, UNIQUE_CHECKS=0;
4      SET @OLD_FOREIGN_KEY_CHECKS=@@FOREIGN_KEY_CHECKS, FOREIGN_KEY_CHECKS=0;
5      SET @OLD_SQL_MODE=@@SQL_MODE, SQL_MODE='ONLY_FULL_GROUP_BY,STRICT_TRANS_TABLES';
6
7      -----
8      -- Schema nomebd
9      -----
10
11     -----
12     -- Schema nomebd
13     -----
14     CREATE SCHEMA IF NOT EXISTS `nomebd` DEFAULT CHARACTER SET utf8 ;
15     USE `nomebd` ;
16
17     -----
18     -- Table `nomebd`.`pessoa`
19     -----
20     CREATE TABLE IF NOT EXISTS `nomebd`.`pessoa` (
21         `id` INT NOT NULL,
22         `nome` VARCHAR(45) NULL,
23         `sobrenome` VARCHAR(45) NULL,
24         `nivelusuario` INT NULL,
```

Save to File...

Copy to Clipboard

Back

Next

Cancel

[Connection Options](#)[Options](#)[Select Objects](#)[Review SQL Script](#)**[Commit Progress](#)**

Forward Engineering Progress

The following tasks will now be executed. Please monitor the execution.
Press Show Logs to see the execution logs.

- ☒ Connect to DBMS
- ☒ Execute Forward Engineered Script
- ☒ Read Back Changes Made by Server
- ☒ Save Synchronization State

Forward Engineer Finished Successfully

[Show Logs](#)[Back](#)[Close](#)[Cancel](#)

The screenshot shows the MySQL Workbench interface. The top toolbar has a red box around the 'Local instance MySQL80' button. The Navigator panel on the left shows a list of schemas: nomebd, sakila, sys, and world. A right-click context menu is open over the 'sakila' schema, showing options: 'Load Spatial Data', 'Create Schema...', and 'Refresh All'. The Query Editor in the center shows a single line of SQL: '1'. The Context Help panel on the right displays 'Automatic context help current'. The bottom status bar shows 'Ready'.

Local instance MySQL80 x MySQL Model* x EER Diagram x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- nomebd
- sakila
- sys
- world

Clicar com o botão direito

- Load Spatial Data
- Create Schema...
- Refresh All

Query 1 x

SQLAdditions

Limit to 1000 rows

Automatic context help current

Administration Schemas

Information

No object selected

Object Info Session

Output

Action Output

#	Time	Action	Message
---	------	--------	---------

Ready

Banco de Dados criado!

MySQL Workbench

Local instance MySQL80 x MySQL Model* x EER Diagram x

File Edit View Query Database Server Tools Scripts Help

Navigator

SCHEMAS

Filter objects

nomebd

- Tables
 - peessoa
- Views
- Stored Procedures
- Functions

sakila

sys

world

Query 1 x

Limit to 1000 rows

```
select * from pessoa
```

Automatic c

Result Grid

	id	nome	sobrenome	nivelusuario
*	NULL	NULL	NULL	NULL

Form Editor

Administration Schemas

peessoa 2 x Apply Revert Context Help Snippet

Output

Action Output

#	Time	Action
1	10:48:09	select * from pessoa LIMIT 0, 1000

Java-Banco de Dados

1º passo: Fazer o download e instalar o SGBD no seu computador

2º passo: Criar o Banco de Dados no SGBD

3º passo: Fazer o download do JDBC

4º passo: Inserir o JDBC no Eclipse

4º passo: Desenvolver o programa Java no Eclipse para conectar e realizar operações (inserir, atualizar,remover, consultar) no Banco de Dados que foi criado no passo 2.

JDBC

Fazer o Download JDBC:

<https://www.mysql.com/products/connector/>

Está disponível no AVA!

- MySQL HeatWave

- MySQL Enterprise Edition

- Datasheet (PDF)
- Technical Specification
- MySQL Database
- MySQL Document Store
- Oracle Enterprise Manager
- Enterprise Monitor
- Enterprise Backup
- Enterprise HA
- Enterprise Scalability
- Enterprise Authentication
- Enterprise TDE
- Enterprise Encryption
- Enterprise Masking
- Enterprise Firewall

MySQL Connectors

MySQL provides standards-based drivers for JDBC, ODBC, and .Net enabling developers to build database a choice. In addition, a native C library allows developers to embed MySQL directly into their applications.

Developed by MySQL

ADO.NET Driver for MySQL (Connector/NET)

[Download](#)

ODBC Driver for MySQL (Connector/ODBC)

[Download](#)

JDBC Driver for MySQL (Connector/J)

[Download](#)

Node.js Driver for MySQL (Connector/Node.js)

[Download](#)

Download MySQL Connector/J

Python Driver for MySQL (Connector/Python)

[Download](#)

C++ Driver for MySQL (Connector/C++)

[Download](#)

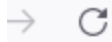
C Driver for MySQL (Connector/C)

[Download](#)

C API for MySQL (mysqlclient)

[Download](#)

These drivers are developed and maintained by the MySQL Community.



https://dev.mysql.com/downloads/connector/j/

MySQL Community Downloads

Connector/J

General Availability (GA) Releases

Archives



Connector/J 8.0.28

Select Operating System:

Platform Independent



Platform Independent (Architecture Independent), Compressed TAR Archive
(mysql-connector-java-8.0.28.tar.gz)

8.0.28

4.0M

Download

MD5: 5f21fdde306a6d643e4c6e8a2ec4b5bd | [Signature](#)

Platform Independent (Architecture Independent), ZIP Archive
(mysql-connector-java-8.0.28.zip)

8.0.28

4.8M

Download

MD5: d19db89de75c46b71c4257badfa1e22b | [Signature](#)

MySQL Community Downloads

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An Oracle Web Account provides you with the following advantages:

- Fast access to MySQL software downloads
- Download technical White Papers and Presentations
- Post messages in the MySQL Discussion Forums
- Report and track bugs in the MySQL bug system

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MySQL.com is using Oracle SSO for authentication. If you already have an Oracle Web account, click the Login link. Otherwise, you can sign up for a free account by clicking the Sign Up link and following the instructions.

No thanks, just start my download.

Arquivo do JDBC

 mysql-connector-java-8.0.28

 src
 build
 CHANGES
 INFO_BIN
 INFO_SRC
 LICENSE
  mysql-connector-java-8.0.28
 README

Java-Banco de Dados

1º passo: Fazer o download e instalar o SGBD no seu computador

2º passo: Criar o Banco de Dados no SGBD

3º passo: Fazer o download do JDBC

4º passo: Inserir o JDBC no Eclipse

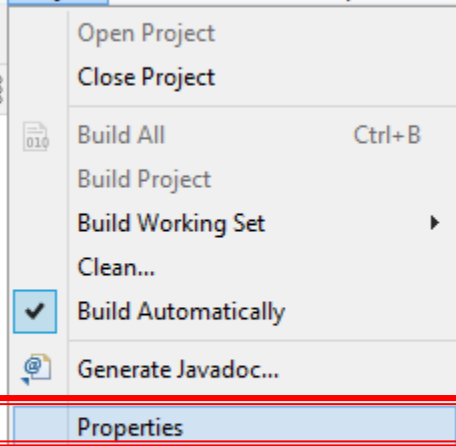
4º passo: Desenvolver o programa Java no Eclipse para conectar e realizar operações (inserir, atualizar,remover, consultar) no Banco de Dados que foi criado no passo 2.

File Edit Source Refactor Navigate Search Project Run Window Help



Package Explorer X

ProjetoBD

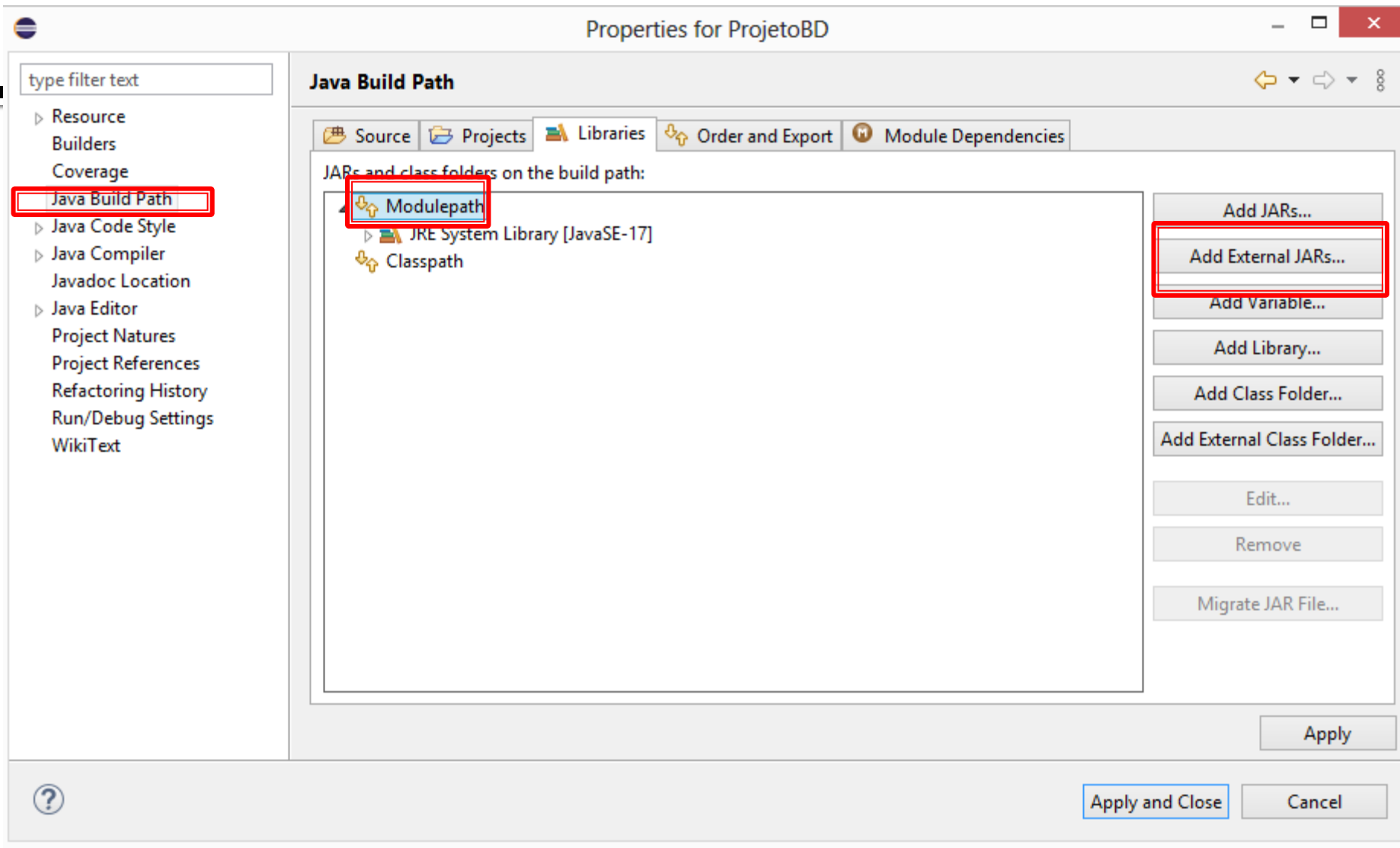


Properties

Problems X @ Javadoc Declaration

0 errors, 1 warning, 0 others

Description	Resource	Path	Location	Type
Warnings (1 item)				



Properties for ProjetoBD

JAR Selection

← → ▾ ↑ 📁 ▶ Aula BD ▶ ▾ ↻ Pesquisar Aula BD 🔍

Organizar ▾

Nova pasta

📁 ▾

📄

?

Nome	Data de modificaç...	Tipo	Tamanho
📁 .metadata	24/04/2022 10:56	Pasta de arquivos	
📁 ProjetoBD	24/04/2022 10:58	Pasta de arquivos	
🔥 mysql-connector-java-8.0.28	16/12/2021 00:25	Executable Jar File	2.419 KB

Nenhuma
visualização
disponível.

Nome: mysql-connector-java-8.0.28 ▾

.jar;.zip ▾

Abrir

Cancelar

Properties for ProjetoBD



type filter text

- ▶ Resource
- Builders
- Coverage
- Java Build Path
- ▶ Java Code Style
- ▶ Java Compiler
- Javadoc Location
- ▶ Java Editor
- Project Natures
- Project References
- Refactoring History
- Run/Debug Settings
- WikiText

Java Build Path



- Source
- Projects
- Libraries
- Order and Export
- Module Dependencies

JARs and class folders on the build path:

- Modulepath
 - mysql-connector-java-8.0.28.jar - C:\Users\angelaabreu\Desktop\Aula BD
 - JRE System Library [JavaSE-17]
 - Classpath

- Add JARs...
- Add External JARs...
- Add Variable...
- Add Library...
- Add Class Folder...
- Add External Class Folder...
- Edit...
- Remove
- Migrate JAR File...

Apply



Apply and Close

Cancel

Java-Banco de Dados

1º passo: Fazer o download e instalar o SGBD no seu computador

2º passo: Criar o Banco de Dados no SGBD

3º passo: Fazer o download e inserir o JDBC no Eclipse

4º passo: Desenvolver o programa Java no Eclipse para conectar e realizar operações (inserir, atualizar, remover, consultar) no Banco de Dados que foi criado no passo 2.

File Edit Source Refactor Navigate Search Project Run Win

Package Explorer X

ProjetoBD

0 errors, 1

Descript

W

New Java Class

Java Class

The use of the default package is discouraged.

Source folder:

Package:

☐ Enclosing type:

Name:

Modifiers: ☒ public ☐ package ☐ private ☐ protected
☐ abstract ☐ final ☐ static
☒ none ☐ sealed ☐ non-sealed ☐ final

Superclass:

Interfaces:

Which method stubs would you like to create?

☒ public static void main(String[] args)
☐ Constructors from superclass
☒ Inherited abstract methods

Do you want to add comments? (Configure templates and default value [here](#))

☐ Generate comments

```
1 import java.sql.Connection;
2 import java.sql.DriverManager;
3 import java.sql.SQLException;
4
5
6 public class ConexaoBancoDeDados
7 {
8
9     private Connection conexao;
10    private final String URL_Bancodedados = "jdbc:mysql://localhost:3306/nomebd";
11    private final String usuario = "root";
12    private final String senha = "admin";
13
14    public ConexaoBancoDeDados()
15    {
16        try { Class.forName("com.mysql.cj.jdbc.Driver");
17              conexao = DriverManager.getConnection(URL_Bancodedados,usuario,senha);
18              System.out.println("Conexao realizada com sucesso!");
19
20
21          } catch (Exception e)
22          {
23              e.printStackTrace();
24              System.out.println("Erro ao acessar o Banco de Dados!");
25          }
26    }
27
28
29    public static void main(String[] args)
30    {
31        ConexaoBancoDeDados conexao = new ConexaoBancoDeDados();
32    }
33
34 }
```

Senha = root

Como fazer um CRUD?

Create

Read

Uppdate

Delete

Inserir Registro no Banco de Dados

```
import java.sql.Statement
```

```
//inserir os dados "Joaquim Saraiva", nível 1 na Tabela Pessoa  
Statement inserir = objBancoDeDados.conexao.createStatement();  
String query = "INSERT INTO pessoa(nome,sobrenome,nivelusuario) VALUES ('JOAQUIM','SARAIVA',1)";  
inserir.execute(query);
```

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
import java.sql.Statement;

public class ConexaoBancoDeDados
{
    private Connection conexao;
    private final String URL_Bancodedados = "jdbc:mysql://localhost:3306/nomebd";
    private final String usuario = "root";
    private final String senha = "admin";

    public ConexaoBancoDeDados()
    {
        try { Class.forName("com.mysql.cj.jdbc.Driver");
              conexao = DriverManager.getConnection(URL_Bancodedados,usuario,senha);
              System.out.println("Conexao realizada com sucesso!");
            } catch (Exception e)
            {
                e.printStackTrace();
                System.out.println("Erro ao acessar o Banco de Dados!");
            }
    }

    public static void main(String[] args) throws SQLException
    {
        ConexaoBancoDeDados objBancoDeDados = new ConexaoBancoDeDados();
        //só irá inserir dados se tiver conseguido acessar o Banco de Dados
        if(objBancoDeDados != null)
        { //inserir os dados "Joaquim Saraiva", nível 1 na Tabela Pessoa
          Statement inserir = objBancoDeDados.conexao.createStatement();
          String query = "INSERT INTO pessoa(nome,sobrenome,nivelusuario) VALUES ('JOAQUIM','SARAIVA',1)";
          inserir.execute(query);
        }
    }
}
```

Local instance MySQL80 x MySQL Model* x EER Diagram x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- nomebd
 - Tables
 - Views
 - Stored Procedures
 - Functions
- sakila
- sys
- world

Query 1 x

Limit to 1000 rows

```
1 select * from pessoa
```

Result Grid

	id	nome	sobrenome	nivelusuario
▶	1	JOAQUIM	SARAIVA	1
*	NULL	NULL	NULL	NULL

Form Editor

Inserir Registro no Banco de Dados

```
import java.sql.PreparedStatement
```

```
//outra opção para fazer o INSERT no banco de dados (obs.: é necessário "import java.sql.PreparedStatement")
```

```
PreparedStatement psInsert = objBancoDeDados.conexao.prepareStatement("INSERT INTO pessoa(nome,sobrenome,nivelusuario) VALUES (?,?,?)");  
psInsert.setString(1, "RENATA");  
psInsert.setString(2, "SILVA");  
psInsert.setInt(3, 3);  
psInsert.execute();
```

```
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.SQLException;
import java.sql.Statement;
import java.sql.PreparedStatement;

public class ConexaoBancoDeDados
{
    private Connection conexao;
    private final String URL_Bancodedados = "jdbc:mysql://localhost:3306/nomebd";
    private final String usuario = "root";
    private final String senha = "admin";

    public ConexaoBancoDeDados()
    {
        try { Class.forName("com.mysql.cj.jdbc.Driver");
            conexao = DriverManager.getConnection(URL_Bancodedados,usuario,senha);
            System.out.println("Conexao realizada com sucesso!");
        } catch (Exception e)
        {
            e.printStackTrace();
            System.out.println("Erro ao acessar o Banco de Dados!");
        }
    }

    public static void main(String[] args) throws SQLException
    {
        ConexaoBancoDeDados objBancoDeDados = new ConexaoBancoDeDados();
        //só irá inserir dados se tiver conseguido acessar o Banco de Dados
        if(objBancoDeDados != null)
        { //inserir os dados "Joaquim Saraiva", nível 1 na Tabela Pessoa
            Statement inserir = objBancoDeDados.conexao.createStatement();
            String query = "INSERT INTO pessoa(nome,sobrenome,nivelusuario) VALUES ('JOAQUIM','SARAIVA',1)";
            inserir.execute(query);

            //outra opção para fazer o INSERT no banco de dados (obs.: é necessário "import java.sql.PreparedStatement")

            PreparedStatement psInsert = objBancoDeDados.conexao.prepareStatement("INSERT INTO pessoa(nome,sobrenome,nivelusuario) VALUES (?, ?, ?)");
            psInsert.setString(1, "RENATA");
            psInsert.setString(2, "SILVA");
            psInsert.setInt(3, 3);
            psInsert.execute();
        }
    }
}
```

Local instance MySQL80 x MySQL Model* x EER Diagram x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

nomebd

- Tables
- Views
- Stored Procedures
- Functions

sakila

sys

world

Query 1 x

1 select * from pessoa

Limit to 1000

Result Grid

Filter Rows:

Edit:

	id	nome	sobrenome	nivelusuario
▶	1	JOAQUIM	SARAIVA	1
	2	JOAQUIM	SARAIVA	1
	3	JOAQUIM	SARAIVA	1
	4	RENATA	SILVA	3
*	NULL	NULL	NULL	NULL

Administration Schemas

pessoa3 pessoa4 x Apply

Leitura de Registros do Banco de Dados

```
import java.sql.ResultSet
```

```
//para fazer consulta no Banco de Dados
```

```
Statement comandoconsulta = objBancoDeDados.conexao.createStatement();
```

```
ResultSet resultadoconsultaBD = comandoconsulta.executeQuery("SELECT * FROM pessoa");
```

```
while(resultadoconsultaBD.next())
```

```
{
```

```
    System.out.println(resultadoconsultaBD.getString("nome") + " " + resultadoconsultaBD.getString("sobrenome") + " Nível: " + resultadoconsultaBD.getString("nivelusuario"));
```

```
}
```

```

1 import java.sql.Connection;
2 import java.sql.DriverManager;
3 import java.sql.SQLException;
4 import java.sql.Statement;
5 import java.sql.PreparedStatement;
6 import java.sql.ResultSet;
7
8 public class ConexaoBancoDeDados
9 {
10     private Connection conexao;
11     private final String URL_Bancodedados = "jdbc:mysql://localhost:3306/nomebd";
12     private final String usuario = "root";
13     private final String senha = "admin";
14
15     public ConexaoBancoDeDados()
16     {
17         try { Class.forName("com.mysql.cj.jdbc.Driver");
18             conexao = DriverManager.getConnection(URL_Bancodedados, usuario, senha);
19             System.out.println("Conexao realizada com sucesso!");
20         } catch (Exception e)
21         {
22             e.printStackTrace();
23             System.out.println("Erro ao acessar o Banco de Dados!");
24         }
25     }
26
27     public static void main(String[] args) throws SQLException
28     {
29         ConexaoBancoDeDados objBancoDeDados = new ConexaoBancoDeDados();
30         //só irá inserir dados se tiver conseguido acessar o Banco de Dados
31         if(objBancoDeDados != null)
32         { //inserir os dados "Joaquim Saraiva", nível 1 na Tabela Pessoa
33             Statement inserir = objBancoDeDados.conexao.createStatement();
34             String query = "INSERT INTO pessoa(nome,sobrenome,nivelusuario) VALUES ('JOAQUIM','SARAIVA',1)";
35             inserir.execute(query);
36
37             //outra opção para fazer o INSERT no banco de dados (obs.: é necessário "import java.sql.PreparedStatement")
38
39             PreparedStatement psInsert = objBancoDeDados.conexao.prepareStatement("INSERT INTO pessoa(nome,sobrenome,nivelusuario) VALUES (?, ?, ?)");
40             psInsert.setString(1, "RENATA");
41             psInsert.setString(2, "SILVA");
42             psInsert.setInt(3, 3);
43             psInsert.execute();
44
45             //para fazer consulta no Banco de Dados
46             Statement comandoconsulta = objBancoDeDados.conexao.createStatement();
47             ResultSet resultadoconsultaBD = comandoconsulta.executeQuery("SELECT * FROM pessoa");
48
49             while(resultadoconsultaBD.next())
50             {
51                 System.out.println(resultadoconsultaBD.getString("nome") + " " + resultadoconsultaBD.getString("sobrenome") + " Nível: " + resultadoconsultaBD.getString("nivelusuario"));
52             }
53         }
54     }
55 }

```

```

        System.out.println("Conexao realizada com sucesso!");
    } catch (Exception e)
    {
        e.printStackTrace();
        System.out.println("Erro ao acessar o Banco de Dados!");
    }
}

public static void main(String[] args) throws SQLException
{
    ConexaoBancoDeDados objBancoDeDados = new ConexaoBancoDeDados();
    //só irá inserir dados se tiver conseguido acessar o Banco de Dados
    if(objBancoDeDados != null)
    {
        //inserir os dados "Joaquim Saraiva", nível 1 na Tabela Pessoa
        Statement inserir = objBancoDeDados.conexao.createStatement();
        String query = "INSERT INTO pessoa(nome,sobrenome,nivelusuario) VALUES ('JOAQUIM','SARA:
        inserir.execute(query);

        //outra opção para fazer o INSERT no banco de dados (obs.: é necessário "import java.sql

        PreparedStatement psInsert = objBancoDeDados.conexao.prepareStatement("INSERT INTO pesso
        psInsert.setString(1, "RENATA");
        psInsert.setString(2, "SILVA");
        psInsert.setInt(3, 3);
        psInsert.execute();

        //para fazer consulta no Banco de Dados
        Statement comandoconsulta = objBancoDeDados.conexao.createStatement();
        ResultSet resultadoconsultaBD = comandoconsulta.executeQuery("SELECT * FROM pessoa");

        while(resultadoconsultaBD.next())
        {
            System.out.println(resultadoconsultaBD.getString("nome") + " " + resultadoconsultaBD.getString("sobrenome") + " Nível: " + resultadoconsultaBD.

        }
    }
}
}

```

```

<terminated> ConexaoBancoDeDados [Java Application] C:\Program f
Conexao realizada com sucesso!
JOAQUIM SARAIVA Nível: 1
JOAQUIM SARAIVA Nível: 1
JOAQUIM SARAIVA Nível: 1
RENATA SILVA Nível: 3
JOAQUIM SARAIVA Nível: 1
RENATA SILVA Nível: 3
JOAQUIM SARAIVA Nível: 1
RENATA SILVA Nível: 3
JOAQUIM SARAIVA Nível: 1
RENATA SILVA Nível: 3

```

Atualizar e Deletar registros no Banco de Dados

```
//atualizar dados
```

```
Statement comandoupdate = objBancoDeDados.conexao.createStatement();  
comandoupdate.execute("UPDATE pessoa SET nome = 'JOCA' WHERE id = 1");
```

```
//deletar um registro do banco de dados
```

```
Statement comandodelete = objBancoDeDados.conexao.createStatement();  
comandoupdate.execute("DELETE FROM pessoa where id = 2");
```

```
public static void main(String[] args) throws SQLException
```

```
{
```

```
    ConexaoBancoDeDados objBancoDeDados = new ConexaoBancoDeDados();
```

```
    //só irá inserir dados se tiver conseguido acessar o Banco de Dados
```

```
    if(objBancoDeDados != null)
```

```
    { //inserir os dados "Joaquim Saraiva", nível 1 na Tabela Pessoa
```

```
        Statement inserir = objBancoDeDados.conexao.createStatement();
```

```
        String query = "INSERT INTO pessoa(nome,sobrenome,nivelusuario) VALUES ('JOAQUIM','SARAIVA',1)";
```

```
        inserir.execute(query);
```

```
        //outra opção para fazer o INSERT no banco de dados (obs.: é necessário "import java.sql.PreparedStatement"
```

```
        PreparedStatement psInsert = objBancoDeDados.conexao.prepareStatement("INSERT INTO pessoa(nome,sobrenome,nivelusuario) VALUES (?, ?, ?)");
```

```
        psInsert.setString(1, "RENATA");
```

```
        psInsert.setString(2, "SILVA");
```

```
        psInsert.setInt(3, 3);
```

```
        psInsert.execute();
```

```
        //para fazer consulta no Banco de Dados
```

```
        Statement comandoconsulta = objBancoDeDados.conexao.createStatement();
```

```
        ResultSet resultadoconsultaBD = comandoconsulta.executeQuery("SELECT * FROM pessoa");
```

```
        //atualizar dados
```

```
        Statement comandoupdate = objBancoDeDados.conexao.createStatement();
```

```
        comandoupdate.execute("UPDATE pessoa SET nome = 'JOCA' WHERE id = 1");
```

```
        //deletar um registro do banco de dados
```

```
        Statement comandodelete = objBancoDeDados.conexao.createStatement();
```

```
        comandodelete.execute("DELETE FROM pessoa where id = 2");
```

```
        while(resultadoconsultaBD.next())
```

```
        {
```

```
            System.out.println(resultadoconsultaBD.getString("nome") + " " + resultadoconsultaBD.getString("sobrenome") + " Nível: " + resultadoconsultaBD.getString("nivelusuario") + " ");
```

```
        }
```

```
    }
```

```
}
```

Problems @ Javadoc Declaration Console

<terminated> ConexaoBancoDeDados [Java Application] C:\

JOCA SARAIVA Nível: 1

JOAQUIM SARAIVA Nível: 1

RENATA SILVA Nível: 3

JOAQUIM SARAIVA Nível: 1

RENATA SILVA Nível: 3

JOAQUIM SARAIVA Nível: 1

RENATA SILVA Nível: 3

JOAQUIM SARAIVA Nível: 1

RENATA SILVA Nível: 3

JOAQUIM SARAIVA Nível: 1

RENATA SILVA Nível: 3

JOAQUIM SARAIVA Nível: 1

RENATA SILVA Nível: 3

JOAQUIM SARAIVA Nível: 1

RENATA SILVA Nível: 3

JOAQUIM SARAIVA Nível: 1

RENATA SILVA Nível: 3

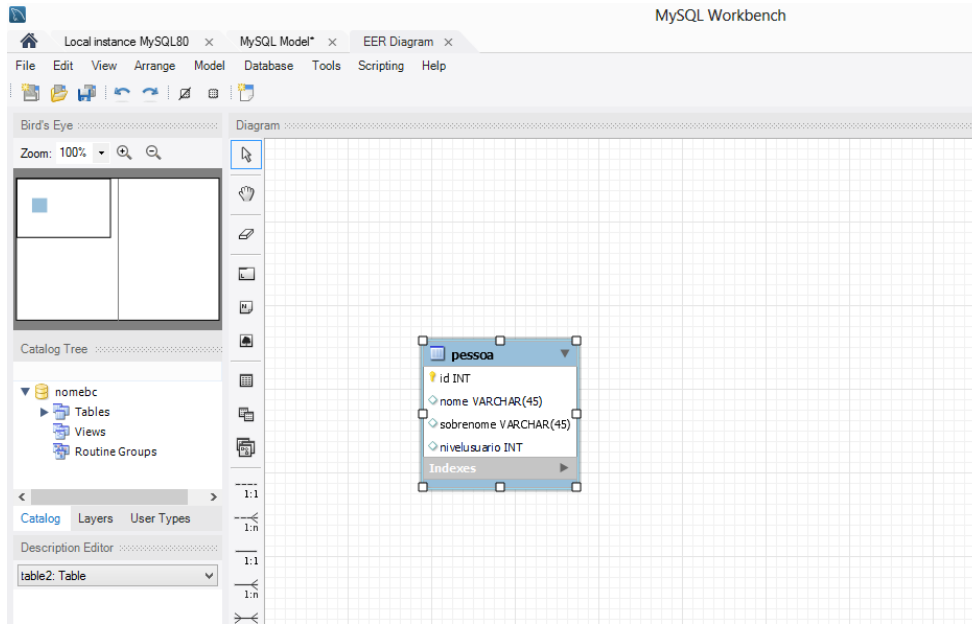
JOAQUIM SARAIVA Nível: 1

RENATA SILVA Nível: 3



Exercícios 😊

Exercício 1



Create
Read
Update
Delete



Slides 63 - 108

Exercício 1

1º passo: Fazer o download e instalar o SGBD no seu computador

2º passo: Criar o Banco de Dados no SGBD

3º passo: Fazer o download do JDBC

4º passo: Inserir o JDBC no Eclipse

4º passo: Desenvolver o programa Java no Eclipse para conectar e realizar operações (inserir, atualizar, remover, consultar) no Banco de Dados que foi criado no passo 2.



Slides 63 - 108



Muito Obrigada!

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